WEEK 7 - HEMATOLOGY • LECTURE 9 • SLIDE COMPANION

Protozoan Parasites of the Blood & Lymphoid Systems

John F. Fisher, MD • one-to-one slide companion for the 85-slide deck

Official Week 7 LOs | 85-slide deck primary | 2 ScholarRx Bricks

Malaria Word doc = supplemental only | Toxoplasmosis = lecture-only extra

Factoid sheet companion

Overview: Official Week 7 Learning Objectives

Malaria. Outline the life cycle; describe pathogenesis; list major clinical manifestations; explain diagnosis; contrast *P. falciparum* with other species in relapse, pathogenesis, and severity; describe antimalarial mechanisms; and summarize treatment/prevention of falciparum versus other species.

Babesiosis. Outline the life cycle, recognize major clinical manifestations, contrast babesiosis with malaria, and summarize diagnosis and treatment.

Chagas disease. Outline the *T. cruzi* life cycle, describe pathogenesis and clinical manifestations, and summarize diagnosis and treatment.

SCHOLARRX BRICKS ASSIGNED FOR THIS LECTURE

Malaria
Protozoa and Ectoparasites

THERE IS NO ONE-TO-ONE WORD DOC FOR THIS LECTURE

The uploaded Word companion is titled **Lecture 10: Malaria** and only develops the malaria portion. Fisher's deck is the primary source and continues through **babesiosis, toxoplasmosis, and Chagas disease**. I use the Word document only where it supplements malaria and never as a substitute for the remaining slides.

OFFICIAL OBJECTIVE VS SLIDE-DECK MISMATCH

Toxoplasmosis is present in Fisher's lecture (Slides 60-67) but is not listed in the official Week 7 protozoan objectives. It is included here because this is a slide companion, but it is labeled as lower official-LO priority. Malaria, babesiosis, and Chagas receive the strongest emphasis.

THE LECTURE'S MASTER COMPARISON

Malaria = mosquito -> liver -> RBC.
Babesia = tick -> RBC only.
Toxoplasma = ingestion -> tissue cysts / intracellular disease.
Chagas = kissing bug feces -> intracellular amastigotes in muscle/nerve -> chronic heart/GI disease.

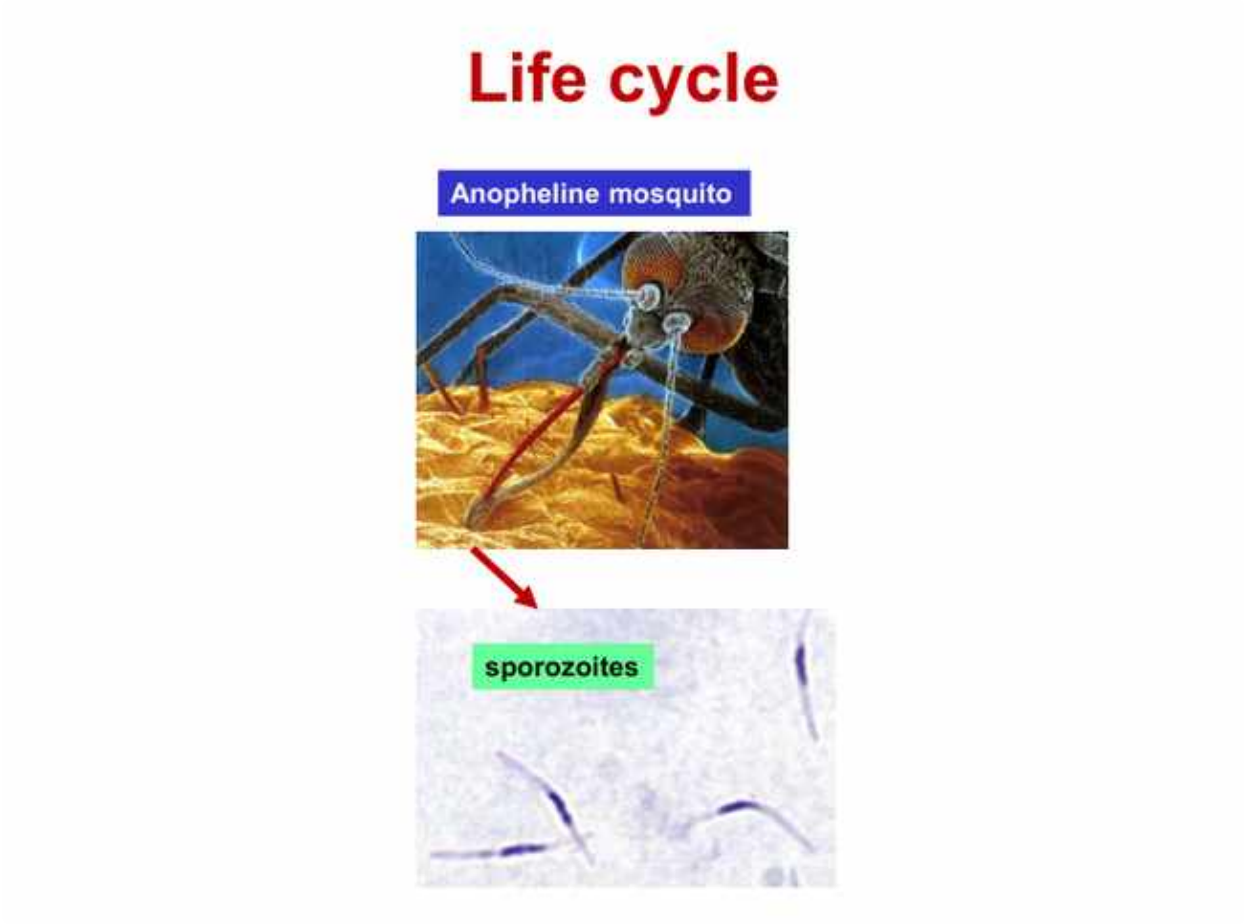
Slides 2-17

Malaria: organism, host-cell preference, and life cycle

Malaria LO a-b

Slides 2-7: organism and epidemiology

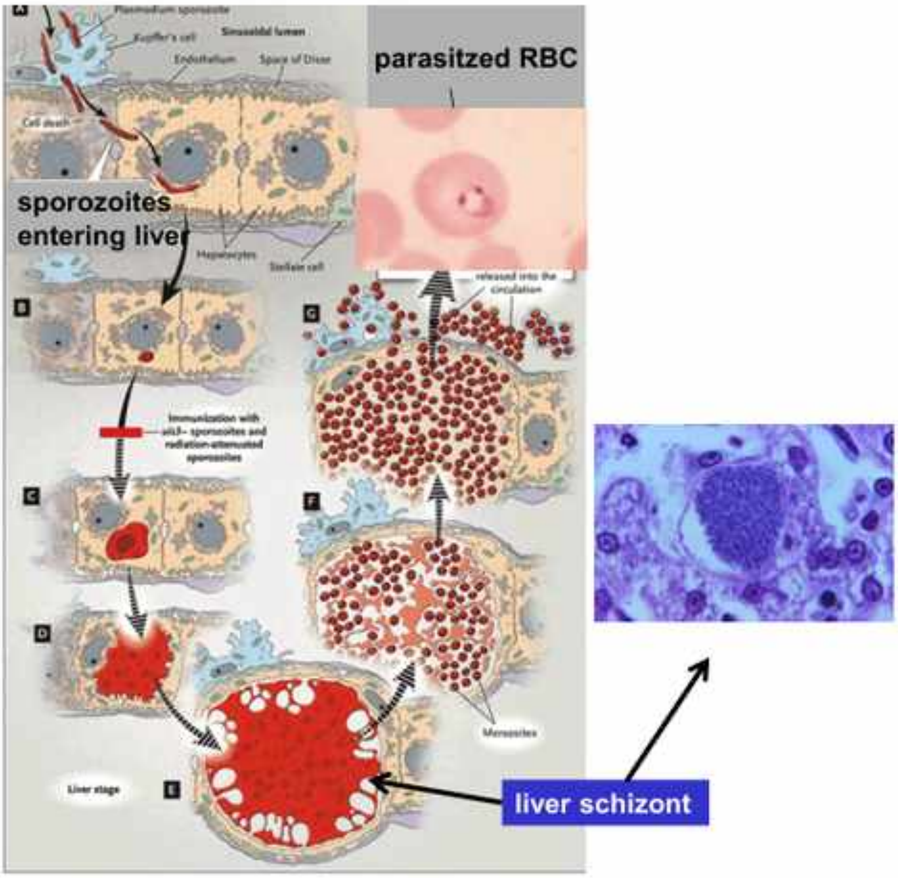
- *Plasmodium* is an **Apicomplexan** protozoan with an apical complex used for host-cell invasion.
- The lecture emphasizes ***P. falciparum*** as the dominant global killer and ***P. vivax*** as another major human species.
- Slide 7 explicitly says **DO NOT MEMORIZE** the individual Anopheles mosquito species. Know the vector category, not those names.



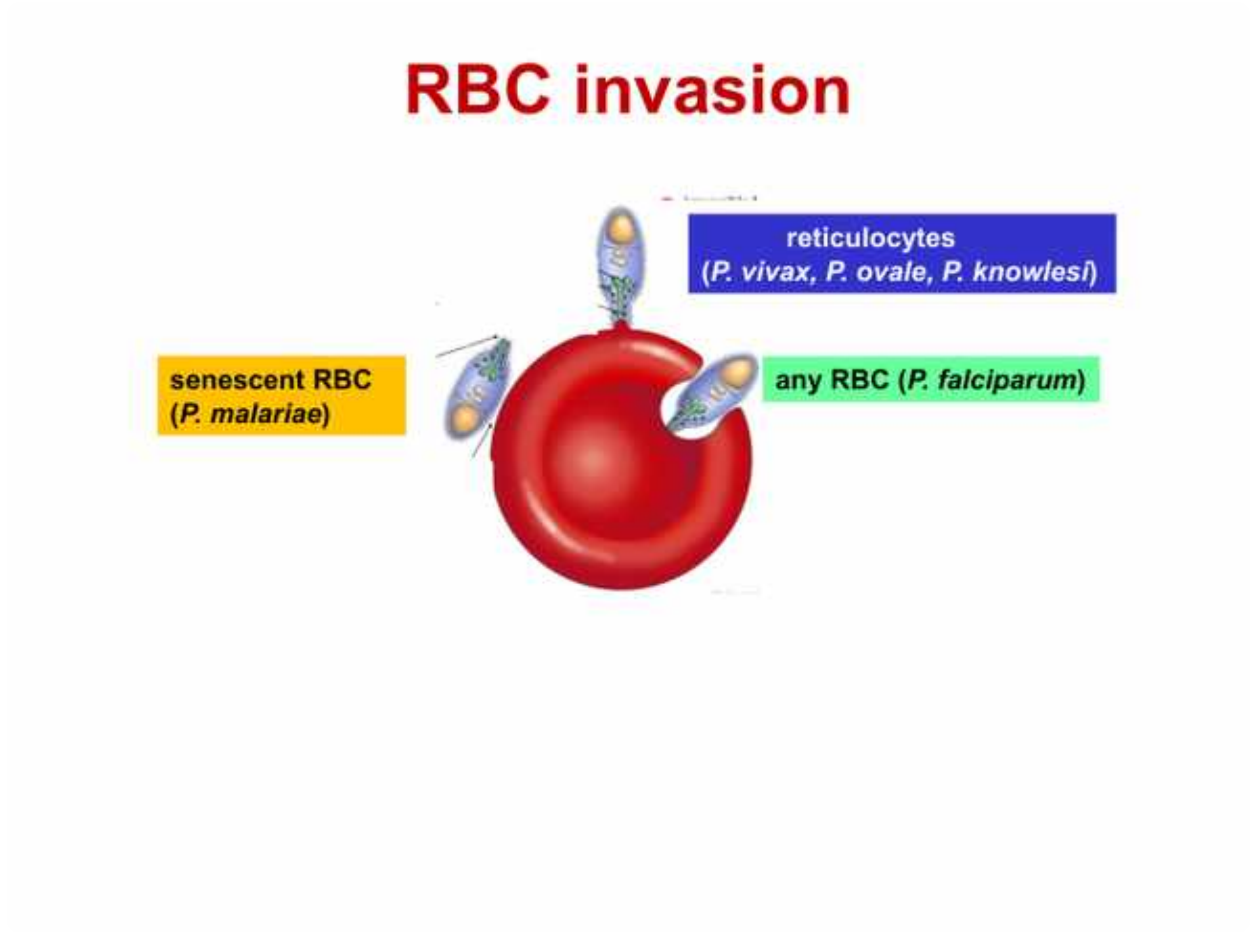
Lecture slide 8 - full slide retained. The infective stage delivered to humans by the female Anopheles mosquito is the sporozoite.

HUMAN LIFE-CYCLE ORDER

Mosquito injects sporozoite -> liver schizont -> merozoites released -> RBC ring trophozoite -> mature trophozoite -> RBC schizont -> RBC rupture -> more merozoites.



Lecture slide 10 - full slide retained. Sporozoites leave blood for hepatocytes, undergo asexual expansion as liver schizonts, then release merozoites into the circulation.

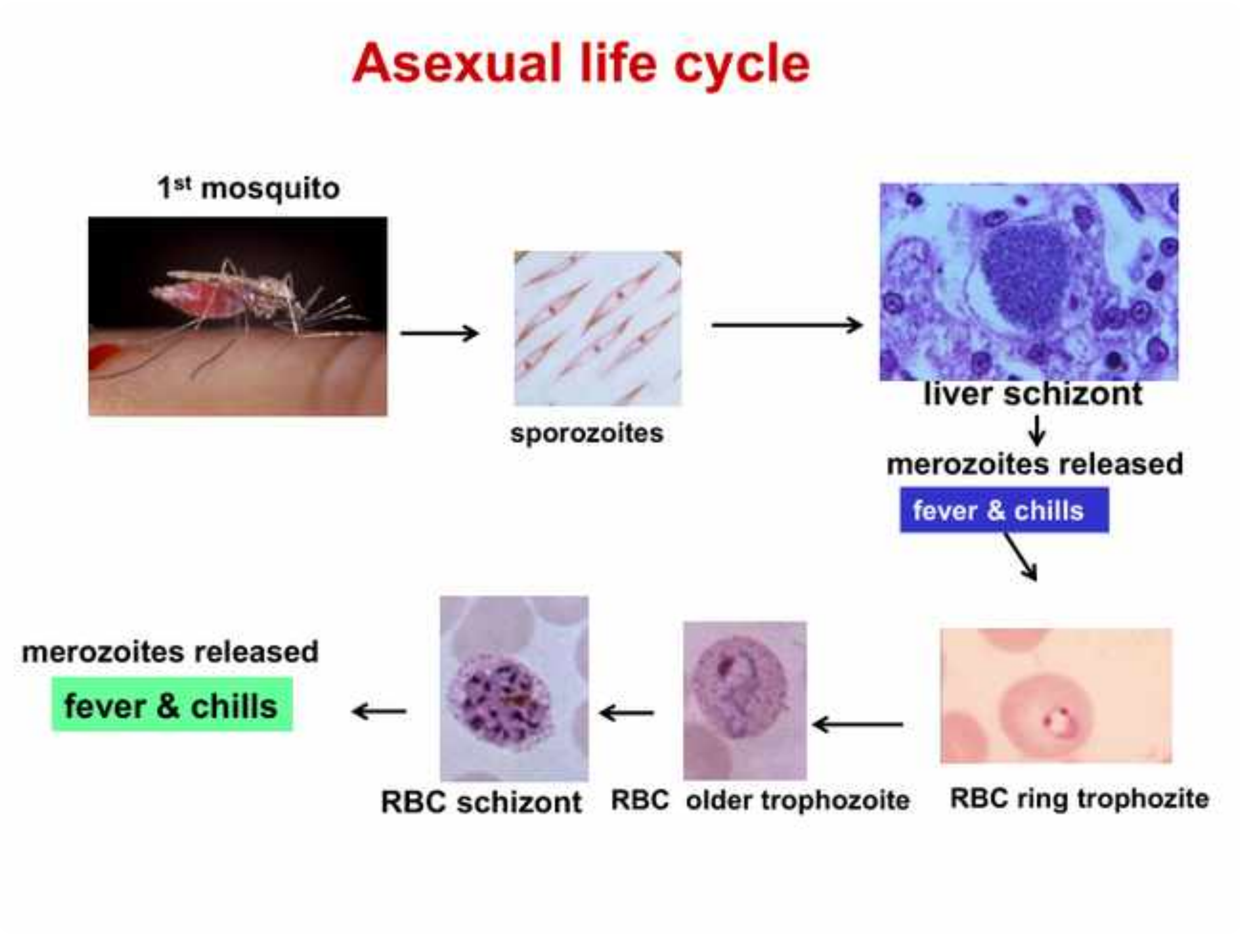


Lecture slide 12 - full slide retained. Fisher distinguishes species by which RBCs they invade. This helps explain maximum parasitemia and disease severity.

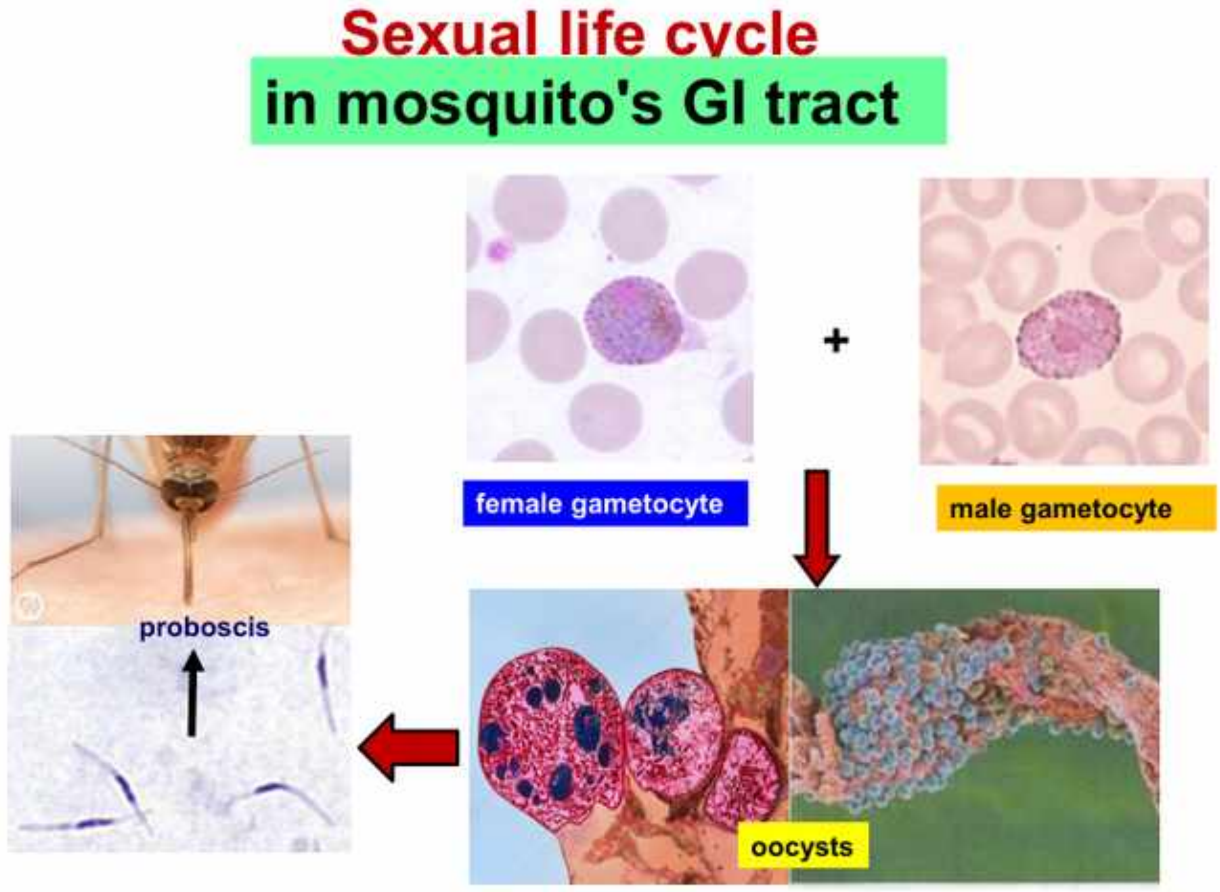
Species	RBC preference in Fisher deck	Why it matters
P. falciparum	**Any RBC**	Can reach high parasitemia -> severe disease.
P. vivax / P. ovale	Reticulocytes	Limited to younger RBC population; relapse is a separate liver issue.
P. knowlesi	Reticulocyte preference on Fisher slide	Fast 24-h replication can still make disease severe.
P. malariae	Senescent RBCs	Lower parasitemia; chronic infection.

Slides 13-15: feeding inside RBCs

The parasite digests hemoglobin for amino acids, generating toxic free heme/hematin. It protects itself by crystallizing this material into **hemozoin (malarial pigment)**. Several drugs exploit this digestive-vacuole pathway later in the deck.



Lecture slide 16 - full slide retained. The asexual human cycle links synchronized RBC rupture to fever and chills. Liver-stage rupture and recurrent RBC schizont rupture release merozoites.



Lecture slide 17 - full slide retained. Gametocytes circulate in human blood, are ingested by another mosquito, and undergo sexual reproduction in the mosquito GI tract before sporozoites ultimately reach the proboscis/salivary apparatus.

SEXUAL REPRODUCTION

Occurs in the mosquito GI tract. Humans carry gametocytes, but fertilization/ sporogony occur in the mosquito.

Slides 18-32

Malaria pathogenesis, fever, host genetics, and severe falciparum disease

Malaria LO b-c-e

Slides 18-19: classic paroxysm

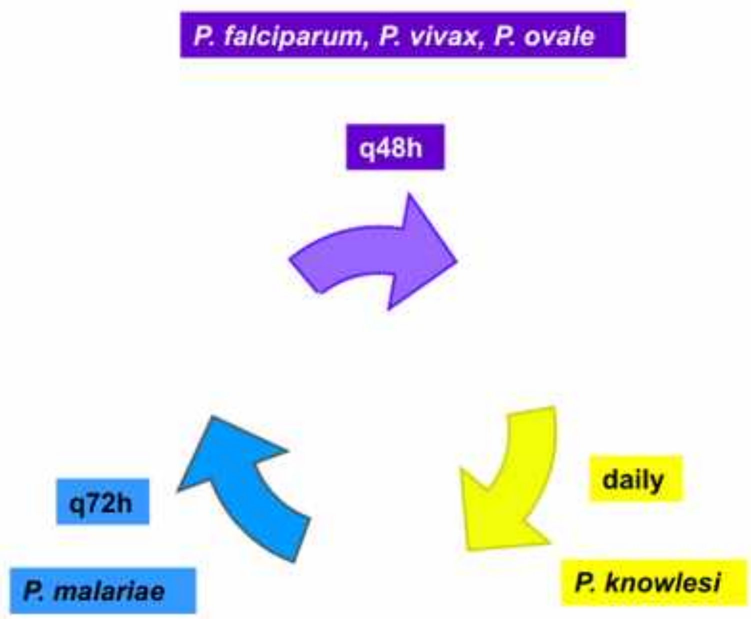
- **Cold stage:** chills/rigors as schizonts rupture and parasite products trigger inflammatory cytokines.
- **Hot stage:** high fever, restlessness, myalgia for hours.
- **Defervescence:** drenching sweats.

Slides 20-24: host resistance

Fisher uses malaria to connect directly back to this week's hematology: sickle trait, hemoglobin C, thalassemia, G6PD deficiency, iron deficiency, hereditary ovalocytosis, and Duffy negativity can alter parasite invasion, growth, or risk of severe disease.

DUFFY CONNECTION

Duffy antigen is a classic receptor used by P. vivax for RBC entry. Fisher emphasizes Duffy-negative populations as strongly protected from classic vivax infection.

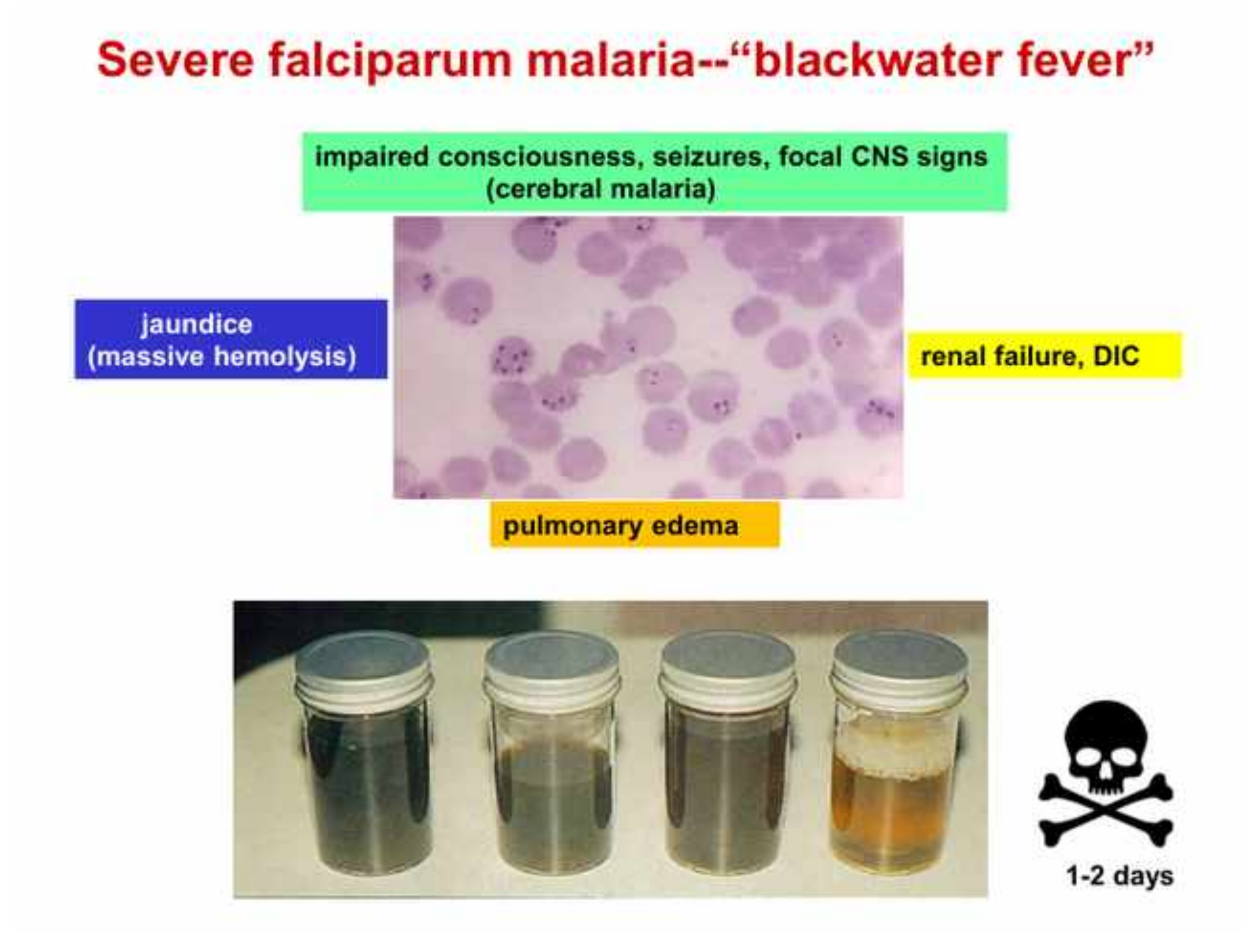


Lecture slide 25 - full slide retained. Fisher's timing map: falciparum/vivax/ovale approximately q48 h, malariae q72 h, knowlesi daily. Real falciparum fever may be irregular rather than perfectly synchronized.

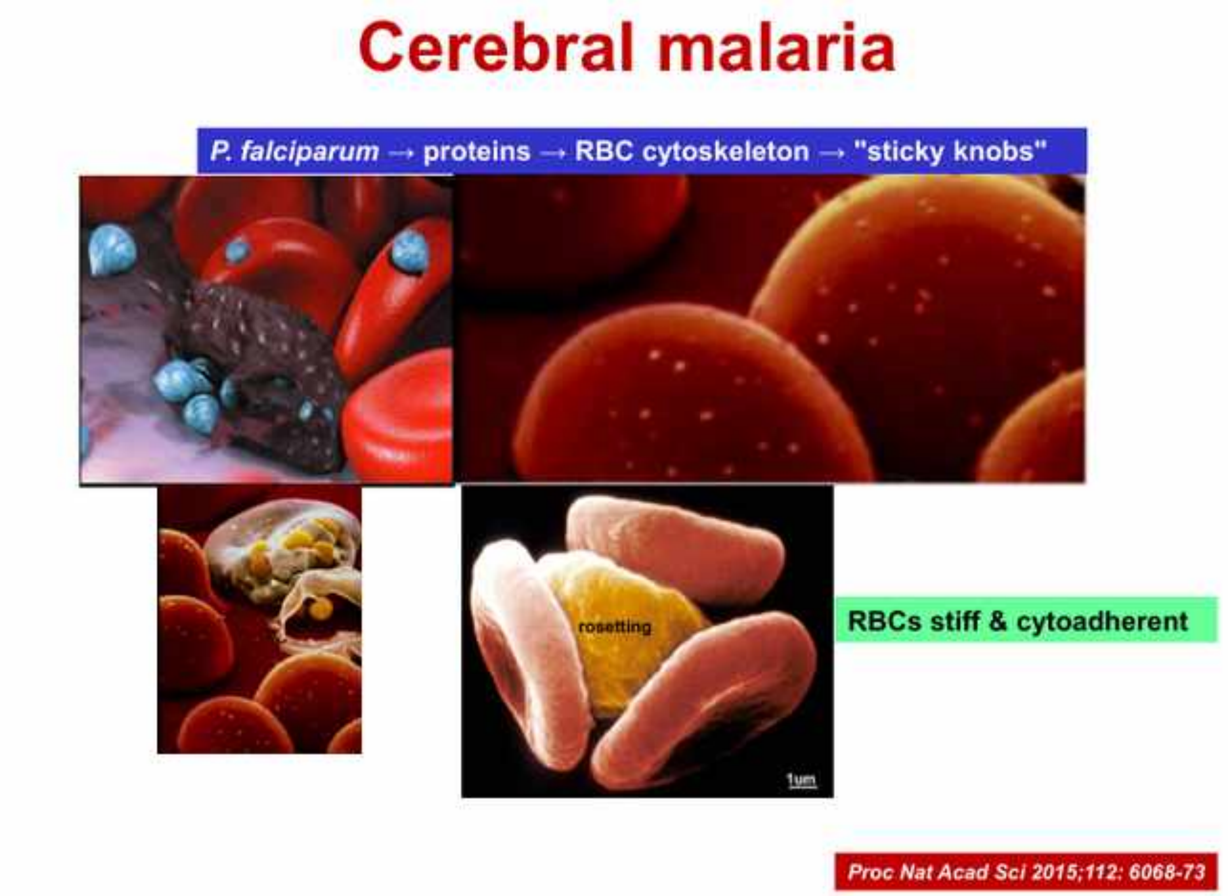
Species	Classic fever rhythm	High-yield label
P. falciparum	~48 h but often irregular clinically	Severe / malignant malaria
P. vivax / ovale	48 h	Tertian
P. malariae	72 h	Quartan
P. knowlesi	24 h	Quotidian / daily

Slides 26-28: severe disease

- Children: impaired consciousness, respiratory distress, jaundice, severe anemia, hypoglycemia, shock/prostration.
- Pregnancy markedly increases susceptibility to severe *P. falciparum* disease in the lecture.
- Severe falciparum complications include **massive hemolysis/ hemoglobinuria, cerebral malaria, pulmonary edema, renal failure, and DIC**.



Lecture slide 28 - full slide retained. Fisher's severe-disease slide links massive hemolysis and jaundice with cerebral, pulmonary, renal, and coagulation complications.



Lecture slide 29 - full slide retained. Falciparum-infected RBCs become stiff and cytoadherent through parasite-induced surface 'sticky knobs,' promoting rosetting and sequestration in small vessels.

WHY FALCIPARUM IS DIFFERENT

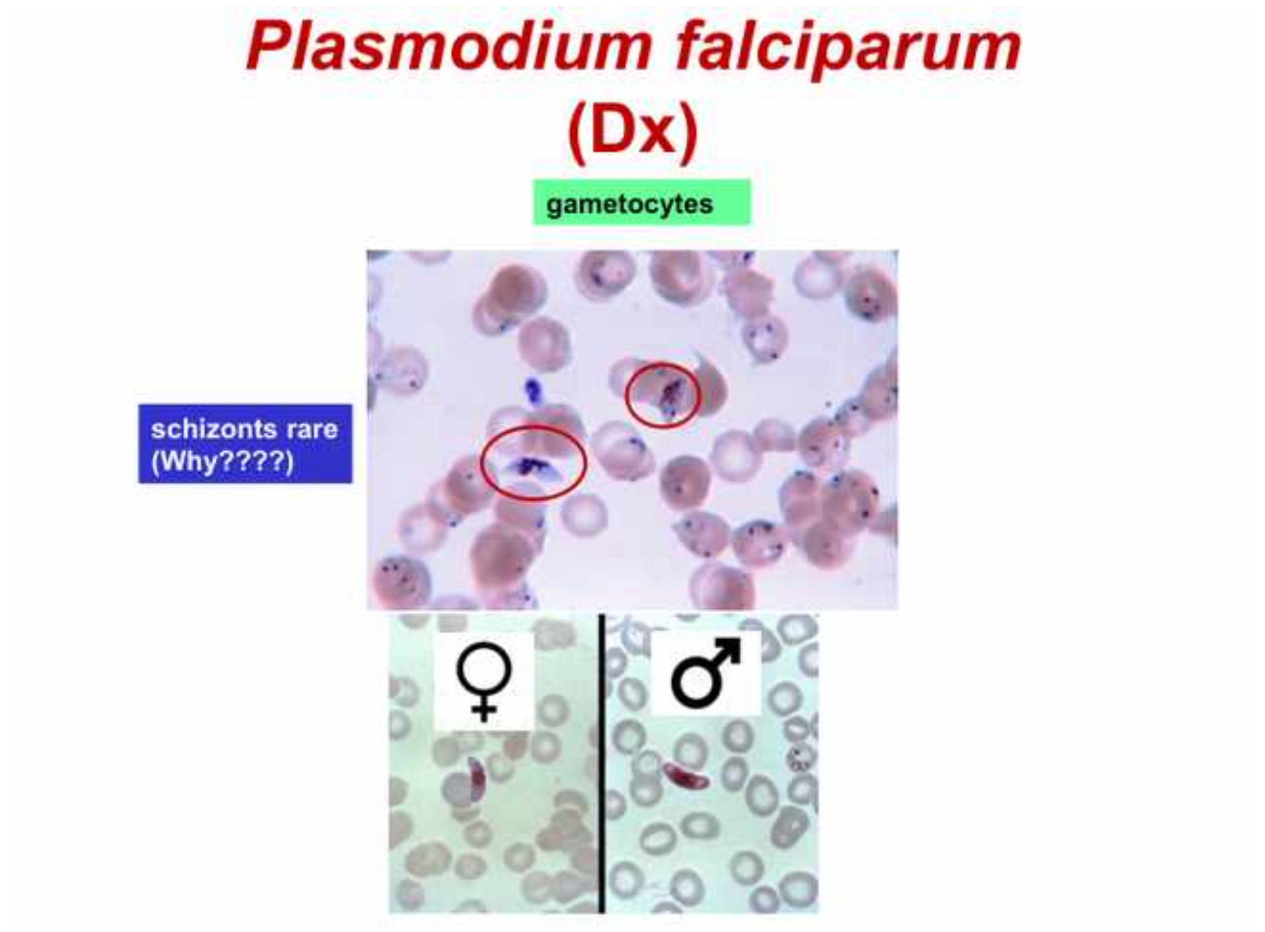
Any RBC + high parasitemia + cytoadherence/sequestration = severe organ disease.

Sequestration also explains why mature trophozoites and schizonts are relatively scarce in peripheral blood.

Slides 33-38

Malaria smear recognition and rapid diagnosis

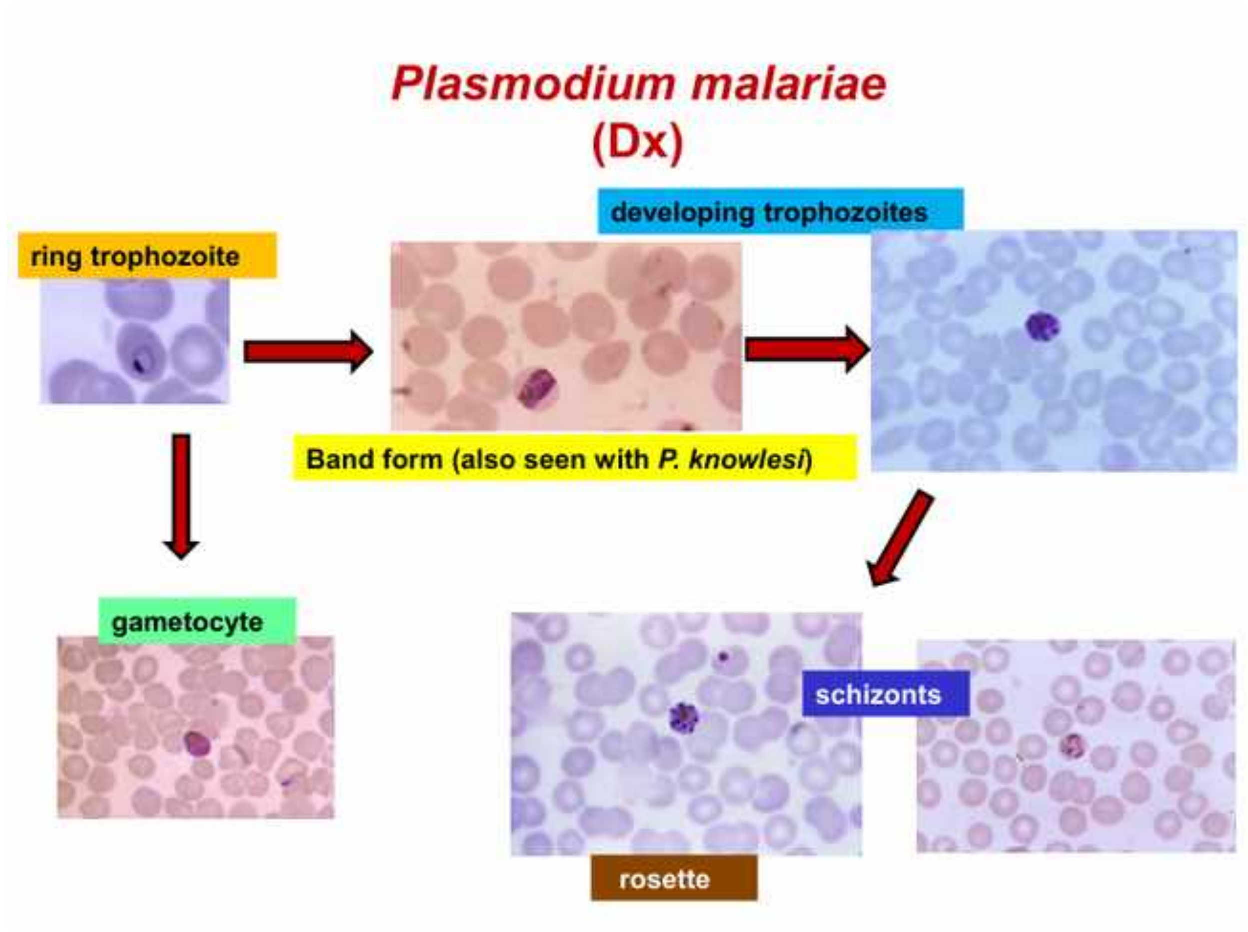
Malaria LO d-e



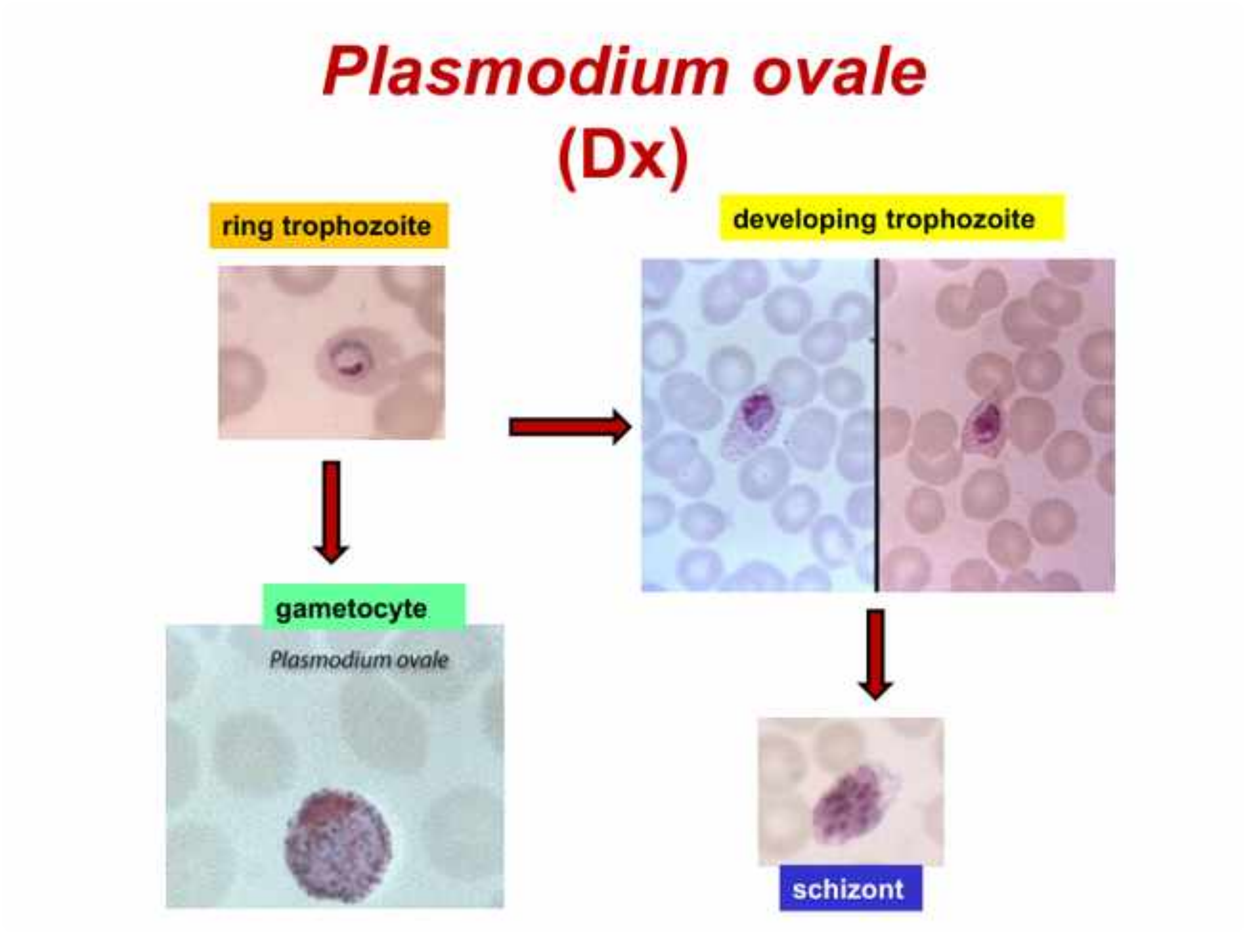
Lecture slide 33 - full slide retained. Falciparum classically shows ring forms and distinctive crescent/banana-shaped gametocytes. Mature schizonts are rare in peripheral blood because of sequestration.

FALCIPARUM SMEAR

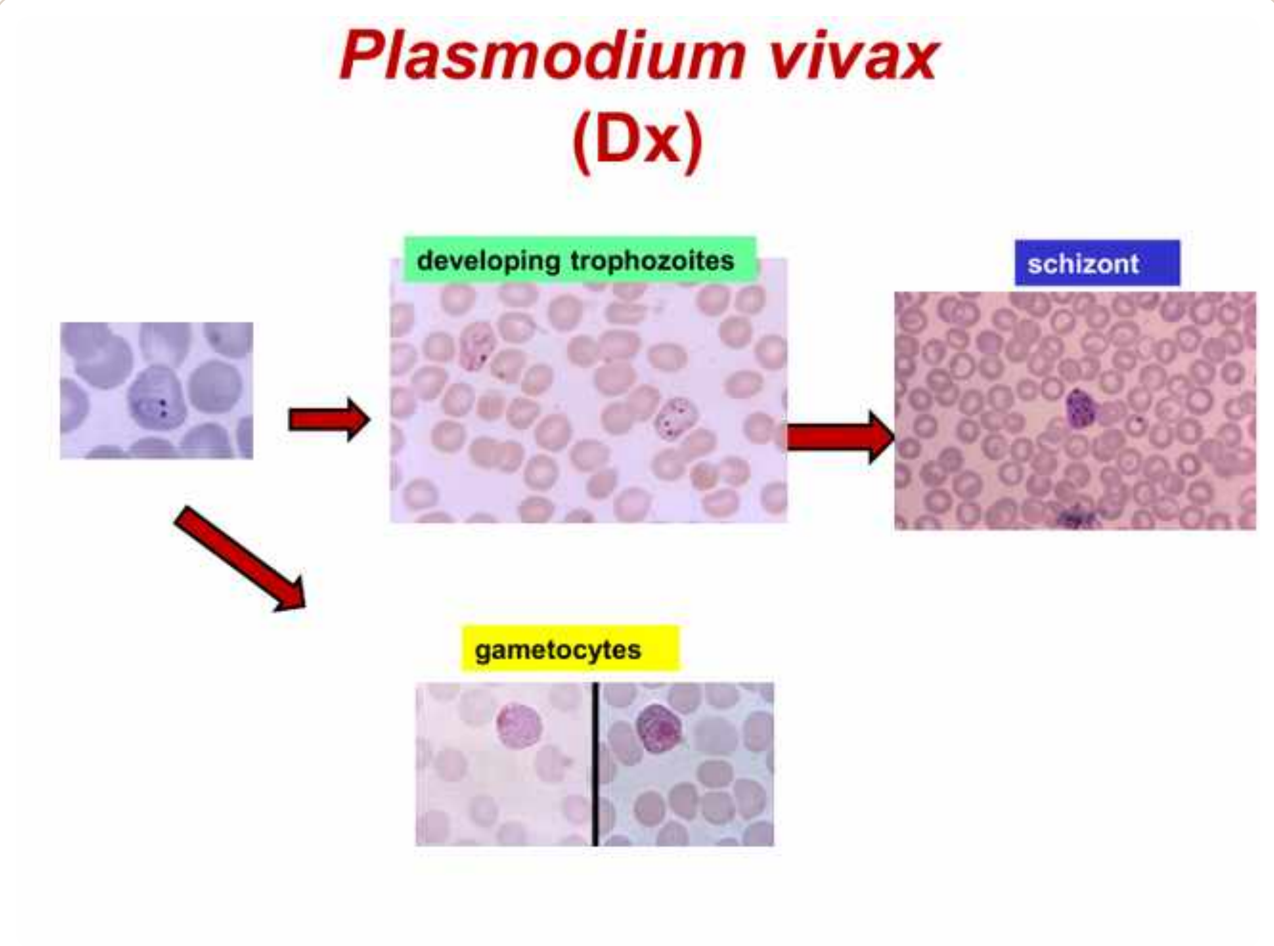
Multiple delicate rings / appliqué forms + crescent gametocytes + few mature schizonts.



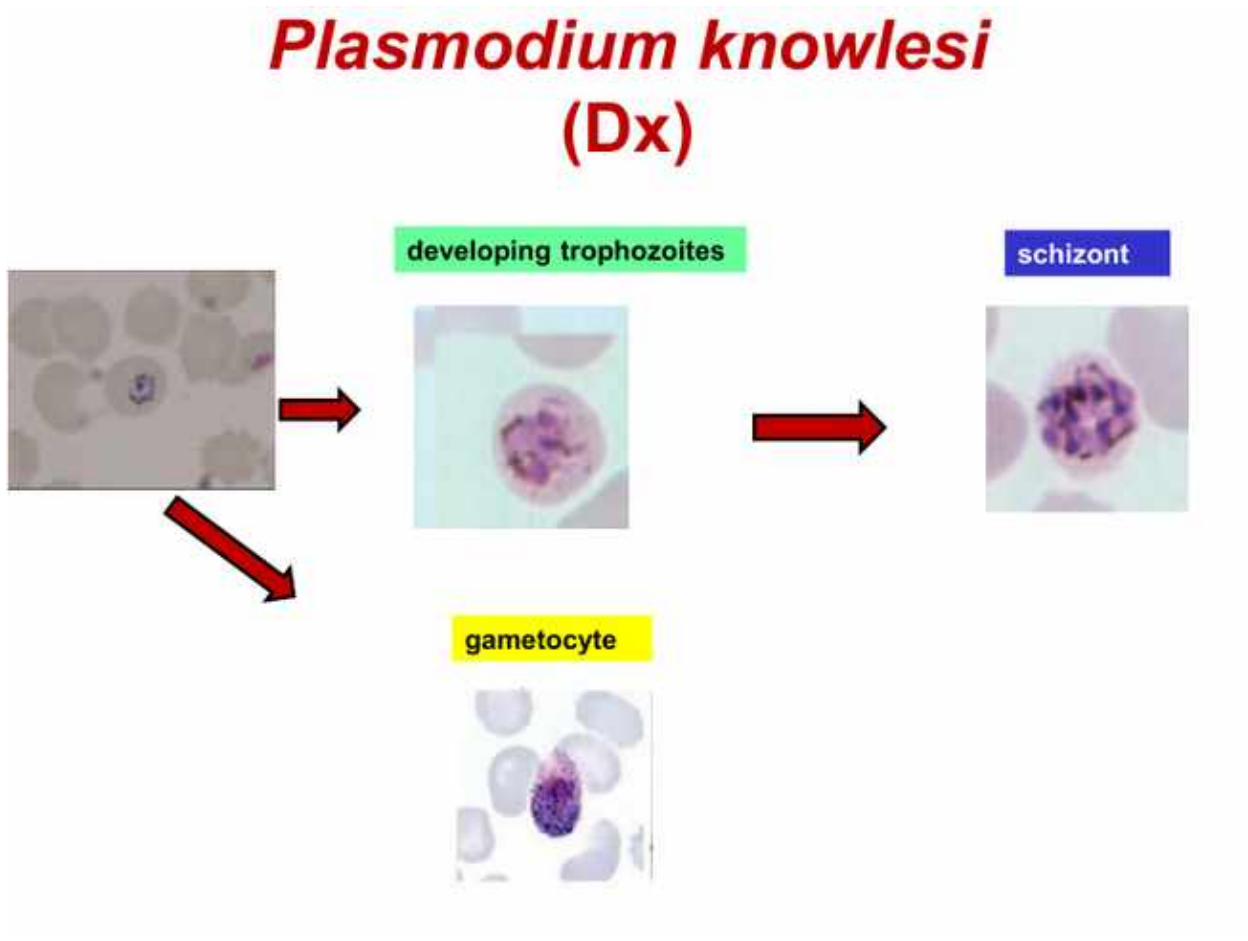
Lecture slide 34 - full slide retained. P. malariae shows band-form trophozoites and rosette-like schizonts; Fisher notes band forms may also be seen with P. knowlesi.



Lecture slide 35 - full slide retained. P. ovale is a relapsing species; the malaria companion further emphasizes oval/fimbriated RBCs and Schüffner-type stippling as board recognition clues.

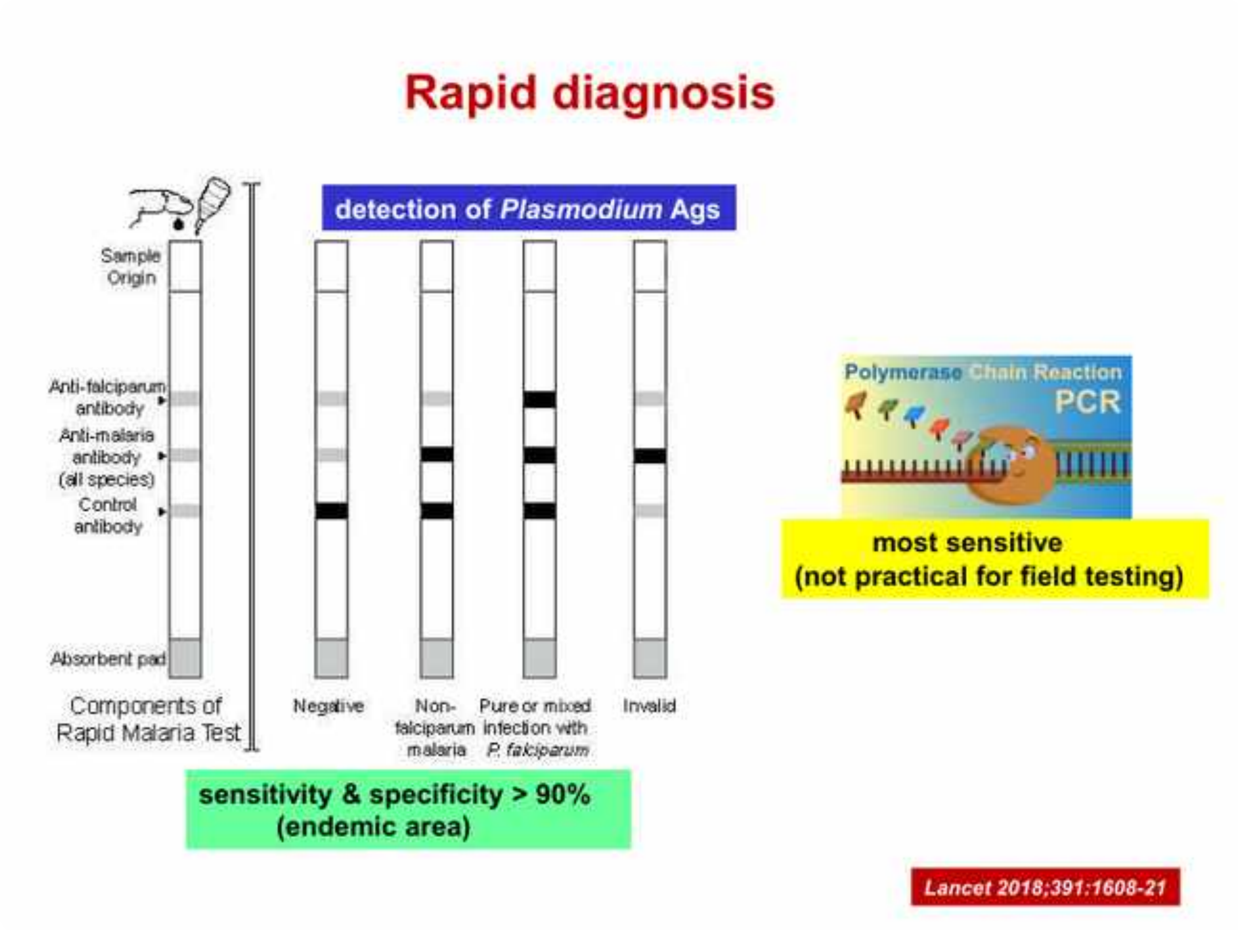


Lecture slide 36 - full slide retained. P. vivax develops in reticulocytes; the supplemental malaria handout emphasizes enlarged RBCs with Schüffner dots.



Lecture slide 37 - full slide retained. Knowlesi can resemble P. malariae morphologically but has a much faster daily replication cycle and may cause severe disease.

Species	Best recognition clue	Relapse?
P. falciparum	Crescent gametocyte, multiple rings, appliqué; mature stages sequester	**No hypnozoite relapse**
P. vivax	Enlarged RBC, Schüffner dots	**Yes - hypnozoites**
P. ovale	Oval/fimbriated RBC, Schüffner-type stippling	**Yes - hypnozoites**
P. malariae	Band form; rosette schizont	No liver hypnozoite relapse
P. knowlesi	May resemble malariae; daily fever	No hypnozoite relapse



Lecture slide 38 - full slide retained. The deck shows antigen-based rapid tests and PCR. The malaria companion supplies the classic microscopy distinction: thick smear for sensitivity/detection, thin smear for species identification.

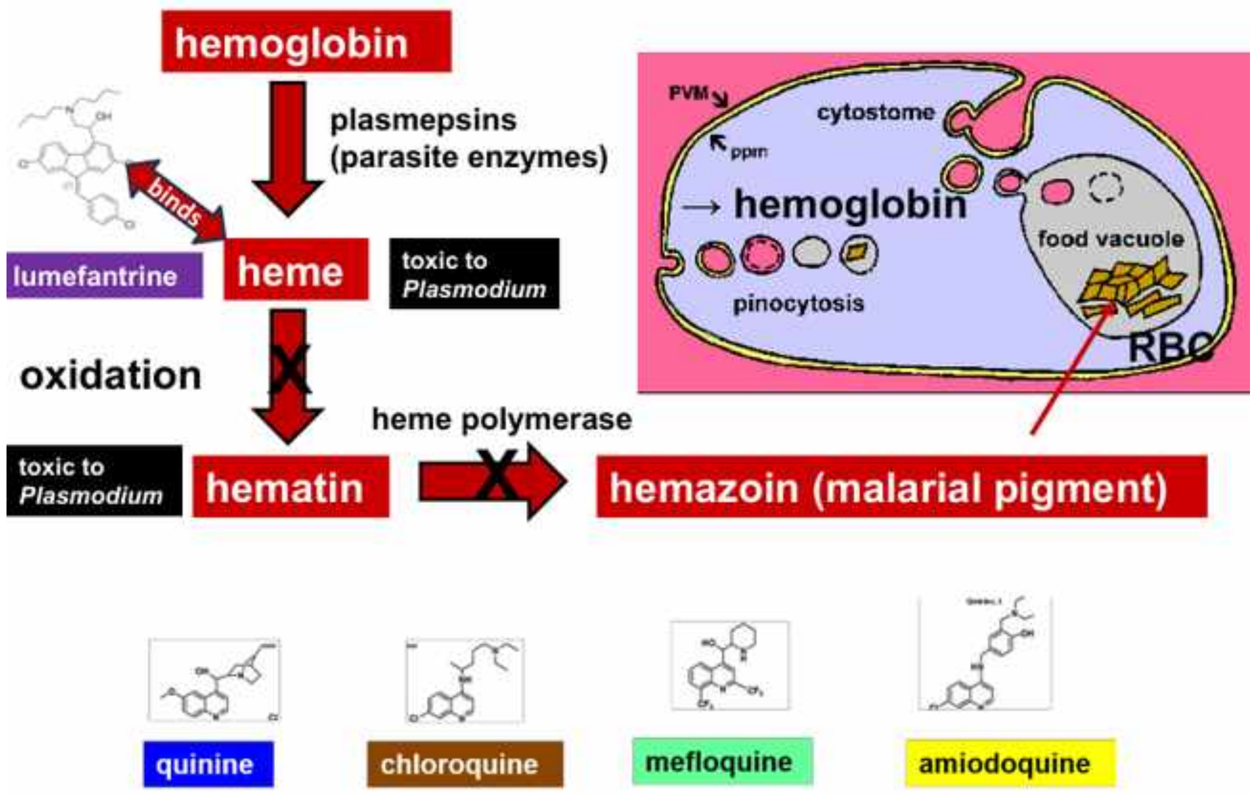
MALARIA WORD-DOC SUPPLEMENT: THICK VS THIN SMEAR

Thick smear: concentrates blood -> sensitive answer to “is malaria present?”
Thin smear: preserves RBC morphology -> species identification and parasitemia assessment.
Giemsa is the classic stain.

Slides 39-49

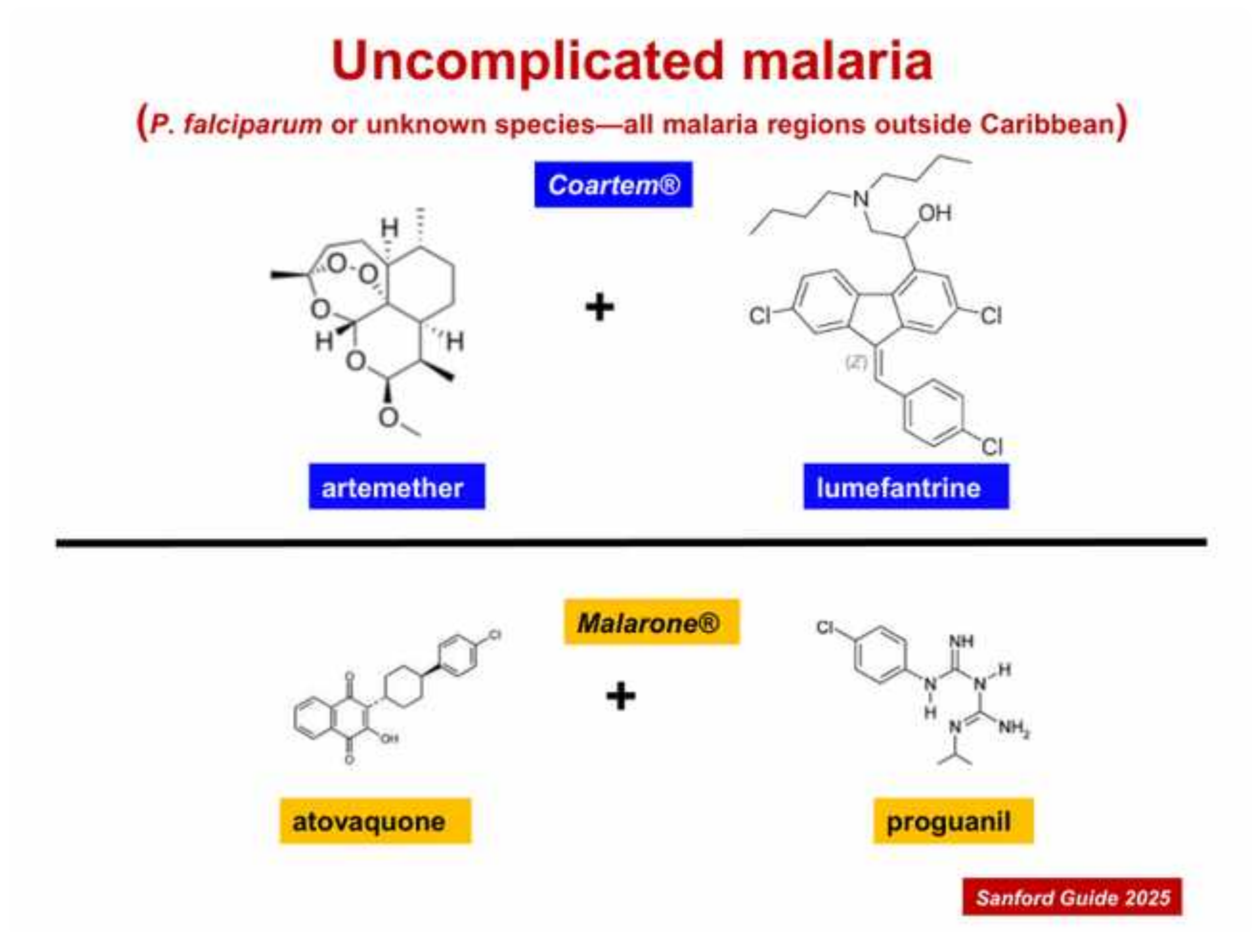
Antimalarial mechanisms, treatment, and prevention

Malaria LO f-g



Lecture slide 39 - full slide retained. Fisher ties quinine/chloroquine/mefloquine/amodiaquine/ lumefantrine to parasite hemoglobin digestion and toxic heme handling inside the food vacuole.

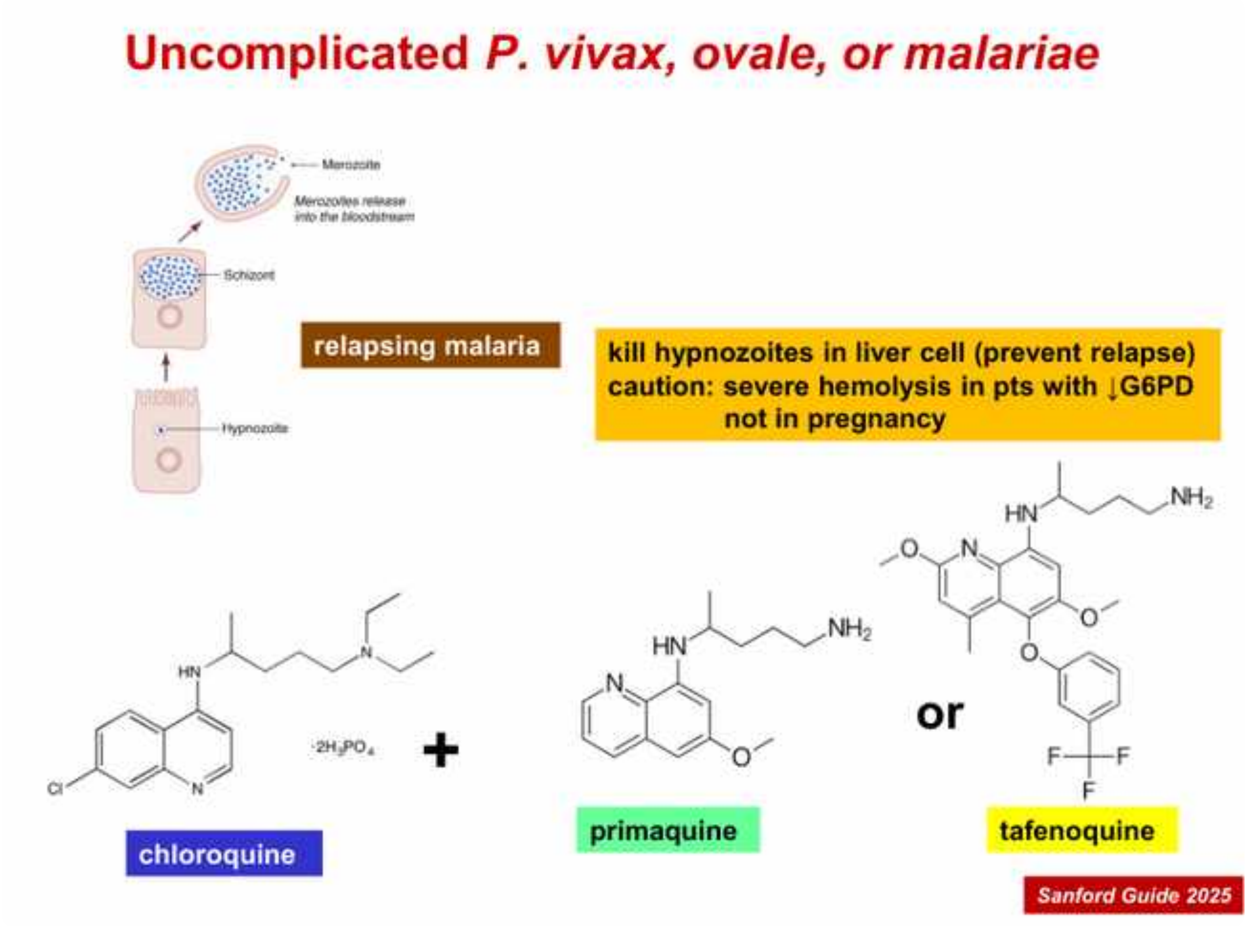
Drug/class in lecture	Mechanistic anchor
Chloroquine / quinine-family heme drugs	Disrupt parasite handling/polymerization of toxic heme in food vacuole.
Artemisinin derivatives	Reactive/free-radical chemistry and multiple parasite targets; artemether/artesunate in deck.
Atovaquone	Structural analogue of coenzyme Q -> mitochondrial electron-transport disruption.
Proguanil	Antifolate pathway; deck diagrams DHFR blockade.
Primaquine / tafenoquine	Liver hypnozoite eradication in relapsing vivax/ovale infection.



Lecture slide 43 - full slide retained. For uncomplicated falciparum or unknown species outside the Caribbean, Fisher lists artemether-lumefantrine or atovaquone-proguanil as treatment options.

SEVERE MALARIA

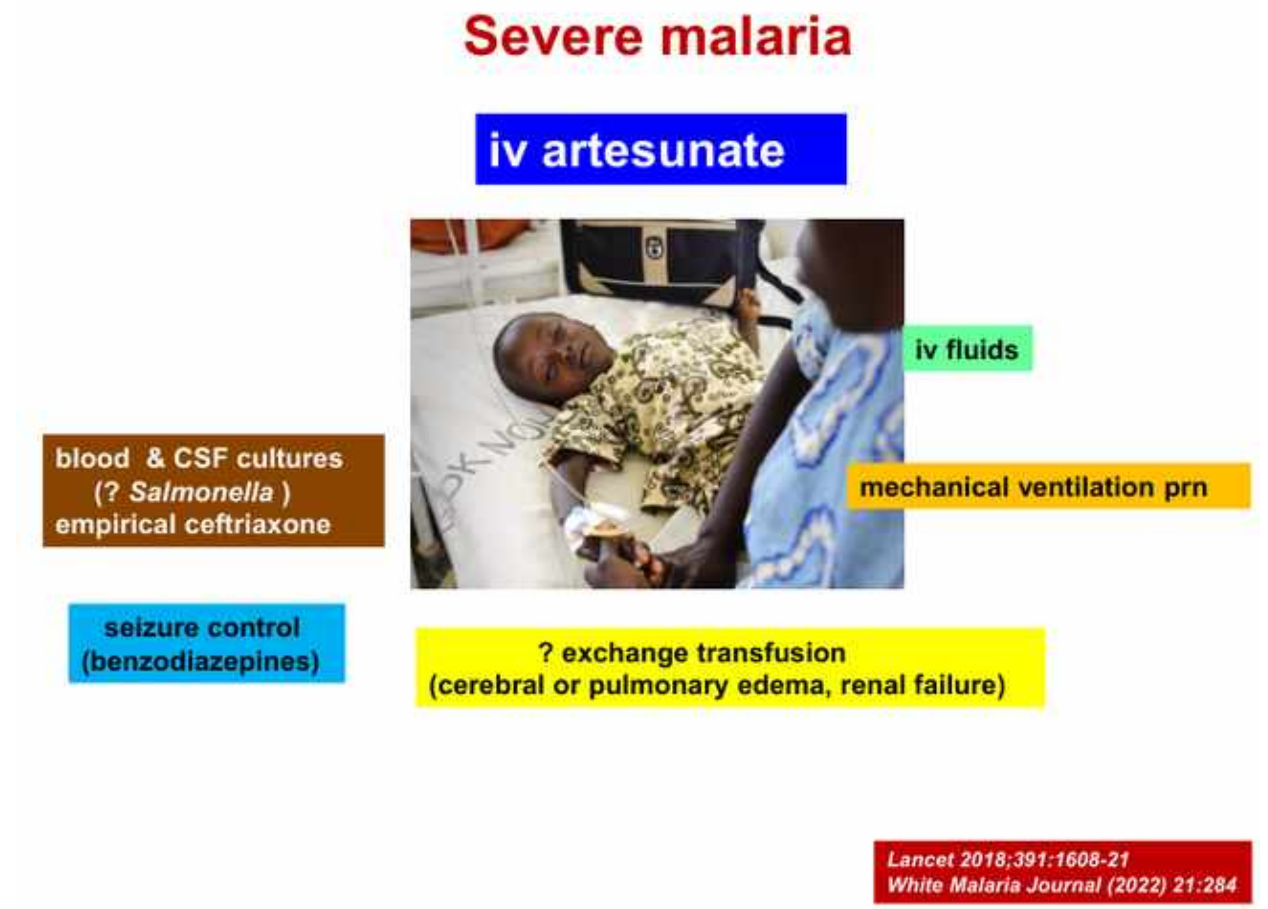
IV artesunate. Then aggressive supportive management for fluid, respiratory, neurologic, renal, and other complications.



Lecture slide 45 - full slide retained. The slide groups vivax/ovale/malariae under uncomplicated malaria, but the radical-cure/hypnozoite concept applies specifically to the relapsing hypnozoite species vivax and ovale. Fisher explicitly warns about G6PD-related hemolysis and pregnancy.

SOURCE NUANCE - DO NOT MISLEARN THIS SLIDE

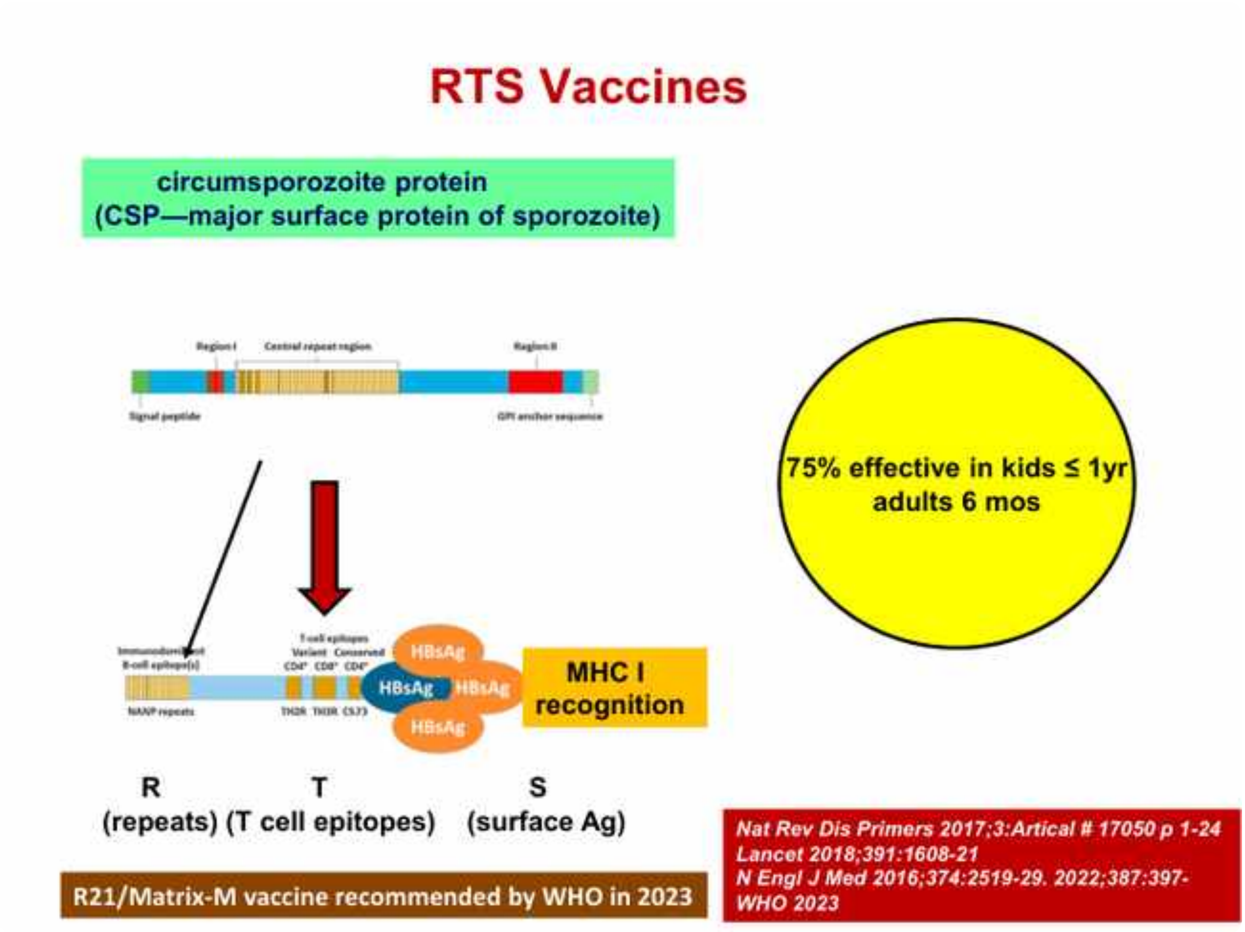
Slide 45's title includes *P. malariae*, but its **primaquine/tafenoquine 'kill hypnozoites' component is for *P. vivax* and *P. ovale***. The uploaded malaria companion explicitly limits hypnozoites and radical cure to vivax/ovale. Always check G6PD status before oxidant 8-aminoquinoline therapy.



Lecture slide 46 - full slide retained. Fisher's severe-malaria slide centers IV artesunate plus supportive care, including seizure/respiratory management and evaluation for concurrent bacterial infection.

Slides 47-49: prevention

The deck reviews whole-sporozoite vaccine concepts, circumsporozoite-protein vaccines, the WHO-recommended R21/Matrix-M vaccine, and experimental monoclonal antibodies against circumsporozoite protein. These are useful conceptual prevention strategies; the official LO simply asks you to summarize prevention rather than memorize every developmental product.



Lecture slide 48 - full slide retained. Fisher highlights circumsporozoite protein as a vaccine target and notes R21/Matrix-M WHO recommendation.

Malaria checkpoint

Can you trace mosquito -> liver -> RBC -> mosquito, distinguish all five major species by one clue, explain falciparum sequestration, and choose **ACT/uncomplicated**, **IV artesunate/severe**, and **hypnozoite radical cure for vivax/ovale**?

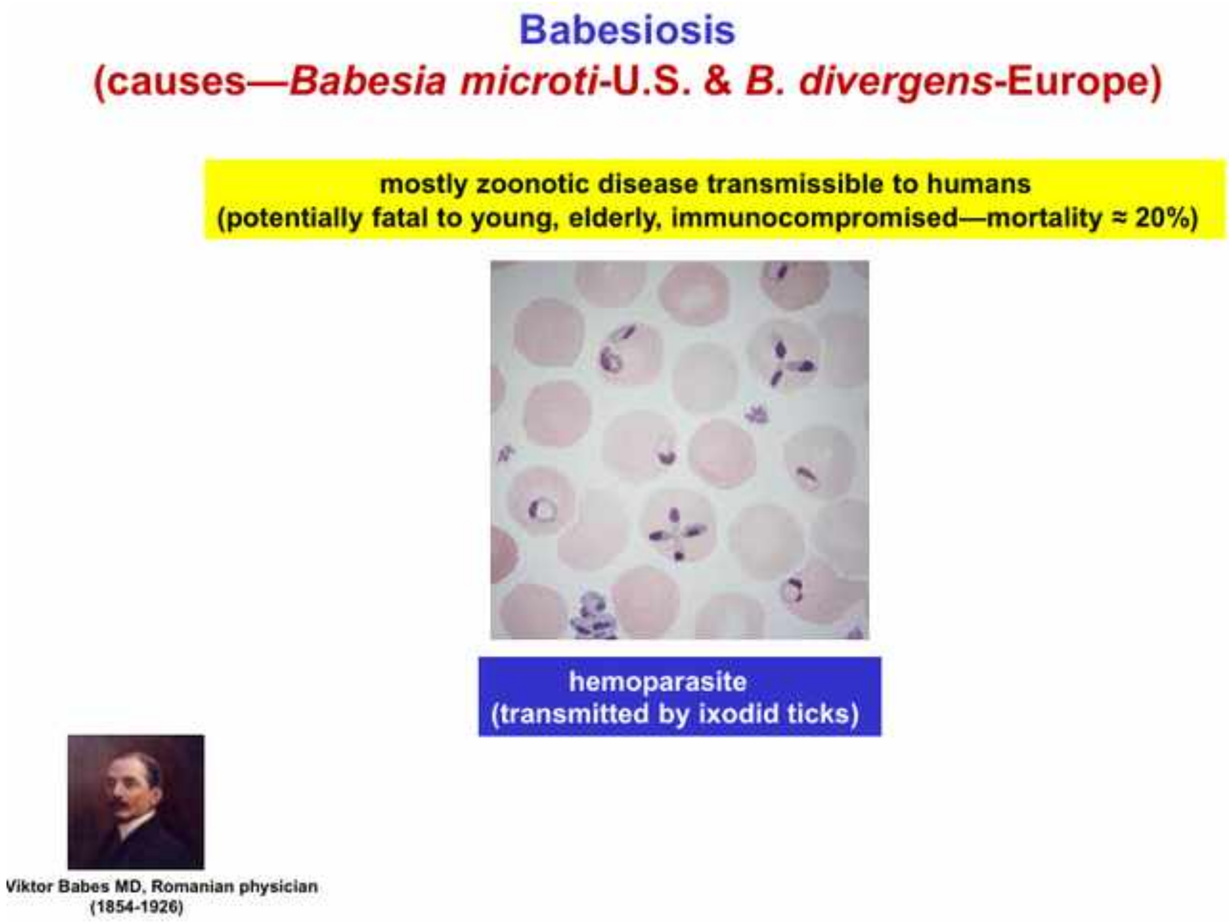
Slides 50-59

Babesiosis: the malaria mimic with no liver stage

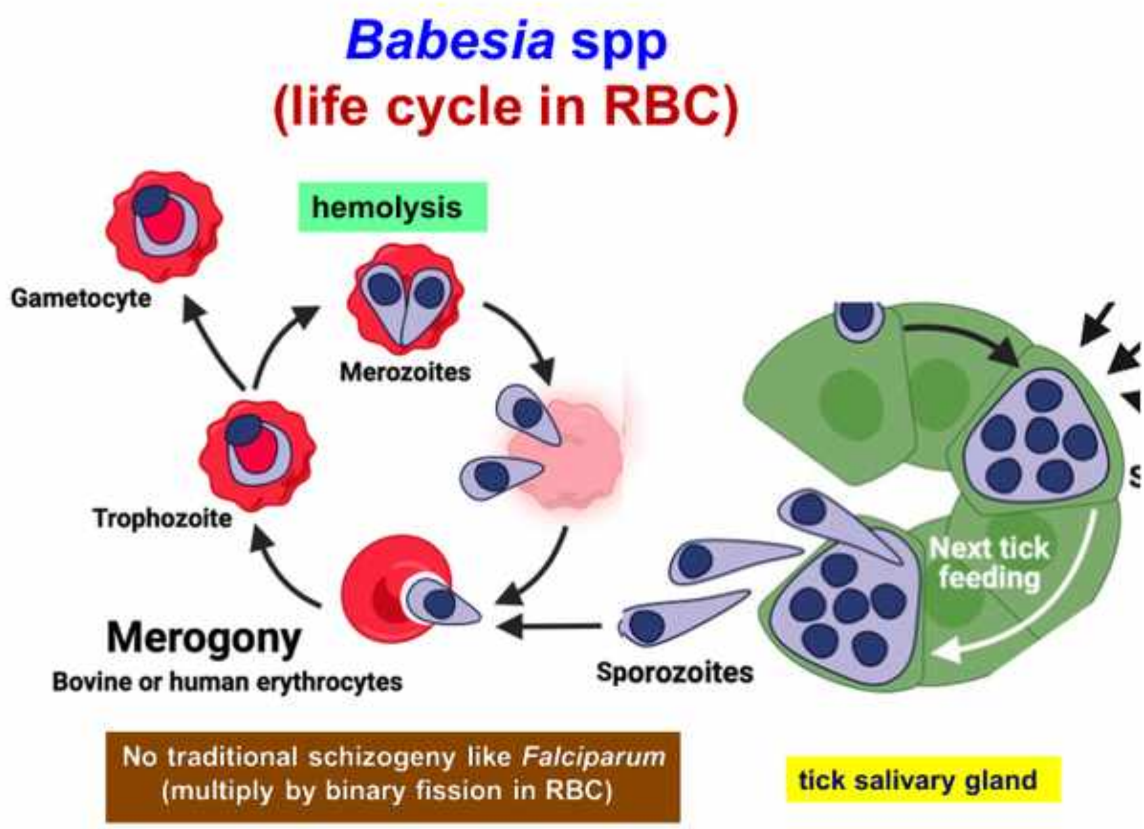
Babesiosis LO

Slides 50-55: organism and cycle

- *Babesia* is also an **Apicomplexan hemoparasite**.
- US disease is classically associated with ***B. microti***; the lecture lists *B. divergens* in Europe.
- Transmission is via **ixodid tick**, often the nymphal stage.
- Unlike malaria, Babesia enters **RBCs directly** and has **no obligatory human liver stage**.
- It is also transfusion-transmissible.



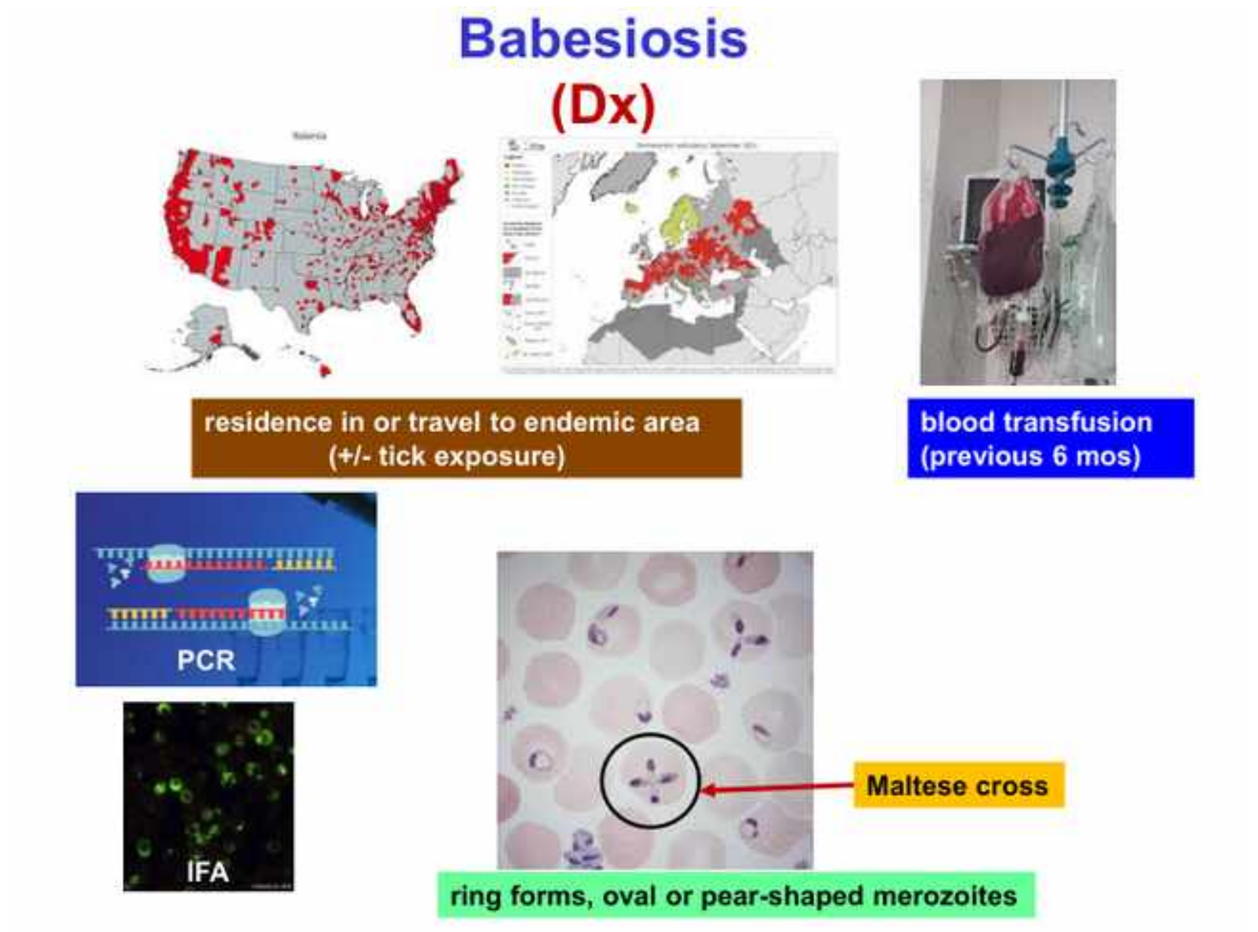
Lecture slide 52 - full slide retained. Fisher emphasizes zoonotic Ixodes transmission and higher mortality in vulnerable hosts.



Lecture slide 54 - full slide retained. Babesia replicates within RBCs and causes hemolysis directly, explaining the malaria-like fever plus hemolytic-anemia picture.

Slides 56-57: clinical disease

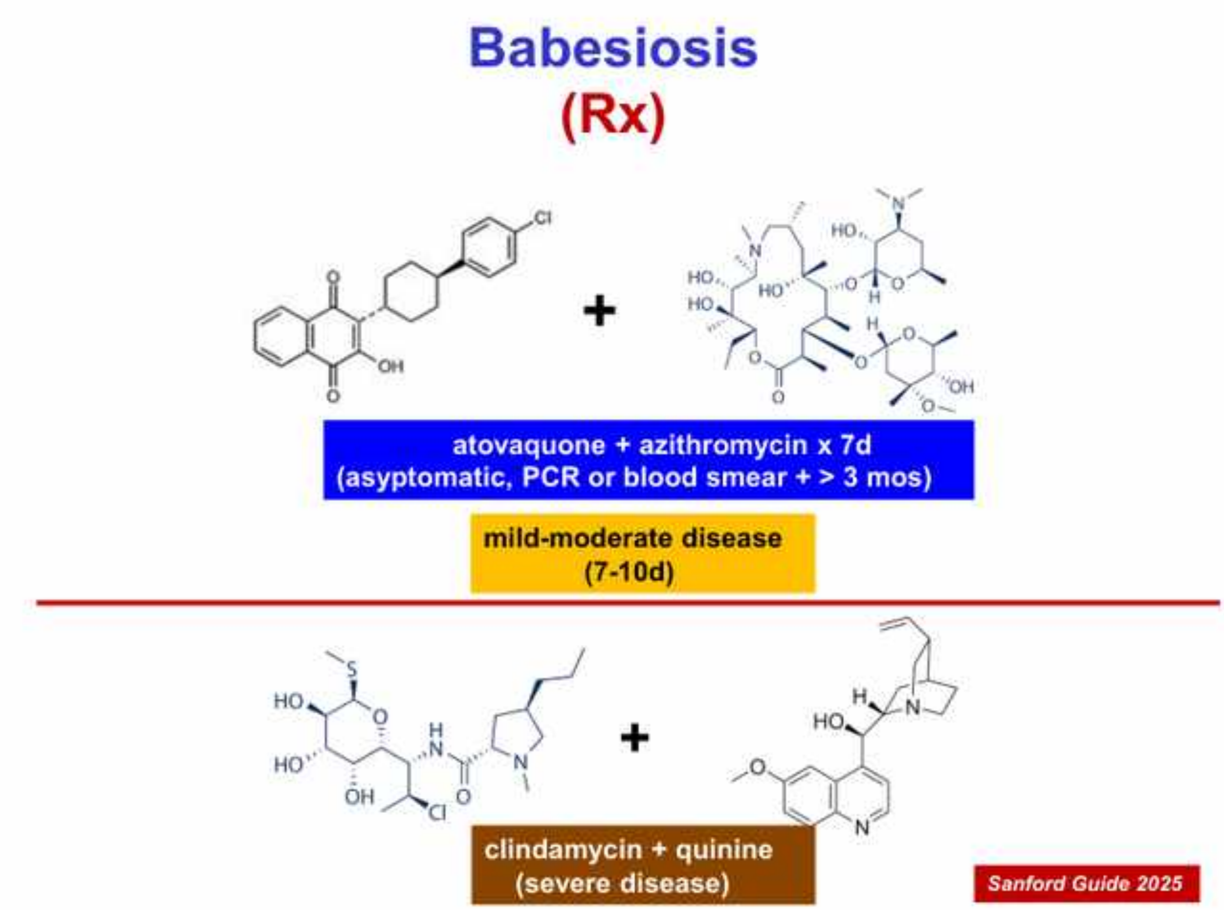
- Incubation around **1-6 weeks** in the deck.
- Fever, chills, headache, myalgia, arthralgia.
- Hemolytic anemia with decreased hemoglobin/platelets and increased bilirubin, LDH, and liver enzymes.
- Severe disease is more likely with older age or immunocompromise and can cause ARDS, CHF, DIC, or renal failure.



Lecture slide 58 - full slide retained. The hallmark smear clue is intraerythrocytic ring/pear forms and especially the Maltese-cross tetrad. PCR and IFA are also listed.

BABESIA COLD SET

Ixodes tick + New England/Northeast-style exposure + hemolytic anemia + Maltese cross + no liver stage.



Lecture slide 59 - full slide retained. Fisher lists atovaquone + azithromycin for mild-moderate disease and clindamycin + quinine for severe disease.

	Malaria	Babesiosis
Vector	Anopheles mosquito	Ixodes tick
Liver phase	**Yes**	**No**
RBC smear	Species-specific rings/ gametocytes/etc.	Ring/pear forms, **Maltese cross**
Transfusion transmission	Possible but not classic teaching anchor	**Explicit lecture clue**
Treatment anchor	Species/severity dependent	Atovaquone + azithro; severe clinda + quinine

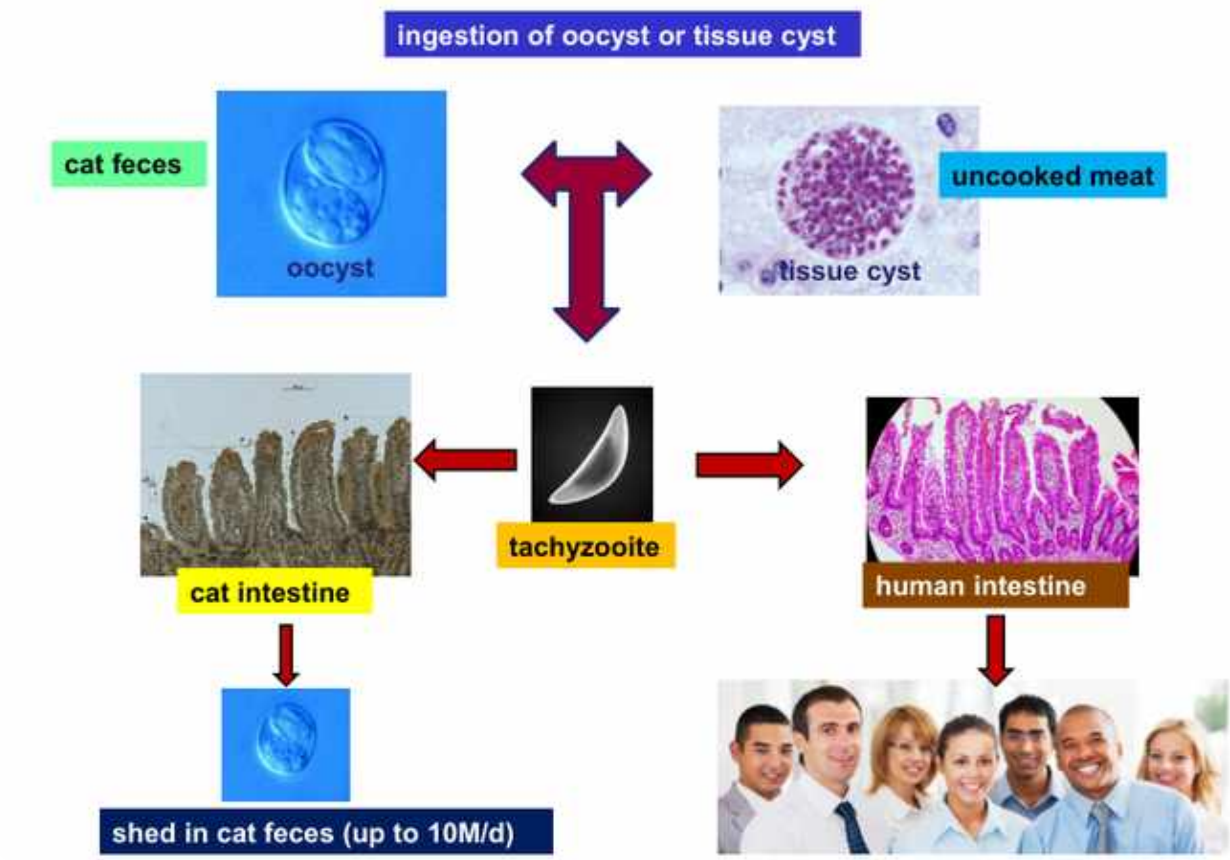
Slides 60-67

Toxoplasmosis - lecture content, lower official-LO priority

Deck extra

WHY THIS IS INCLUDED

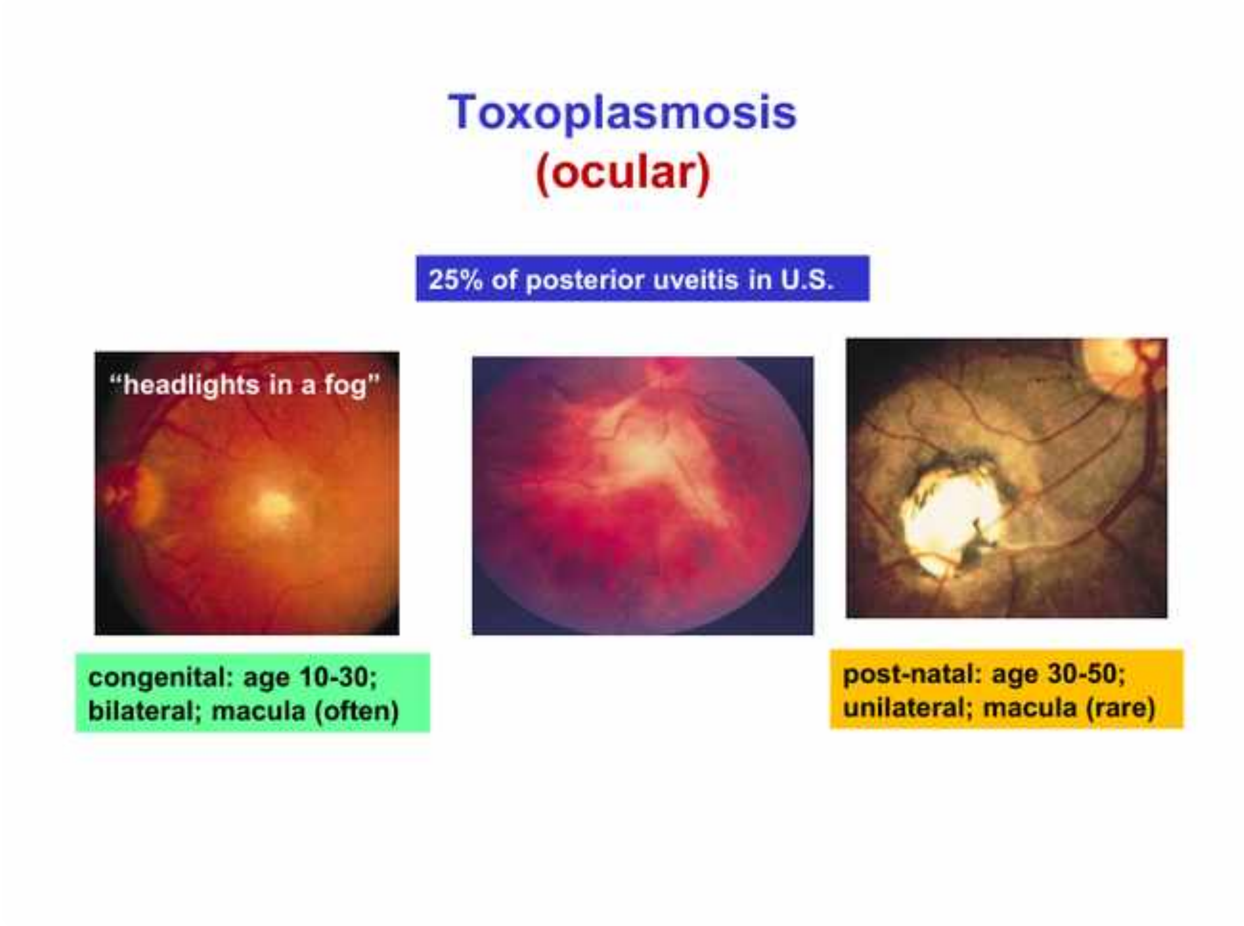
Fisher devotes eight slides to *Toxoplasma gondii*, even though the official Week 7 protozoan objectives do not list toxoplasmosis. Learn the recognition set because it is lecture content, but prioritize malaria/babesiosis/Chagas first when studying directly to the official objectives.



Lecture slide 61 - full slide retained. Humans acquire infection through ingestion of oocysts from cat feces or tissue cysts in undercooked meat. Felids are the definitive host in the lecture.

Slides 62-64: acquired disease spectrum

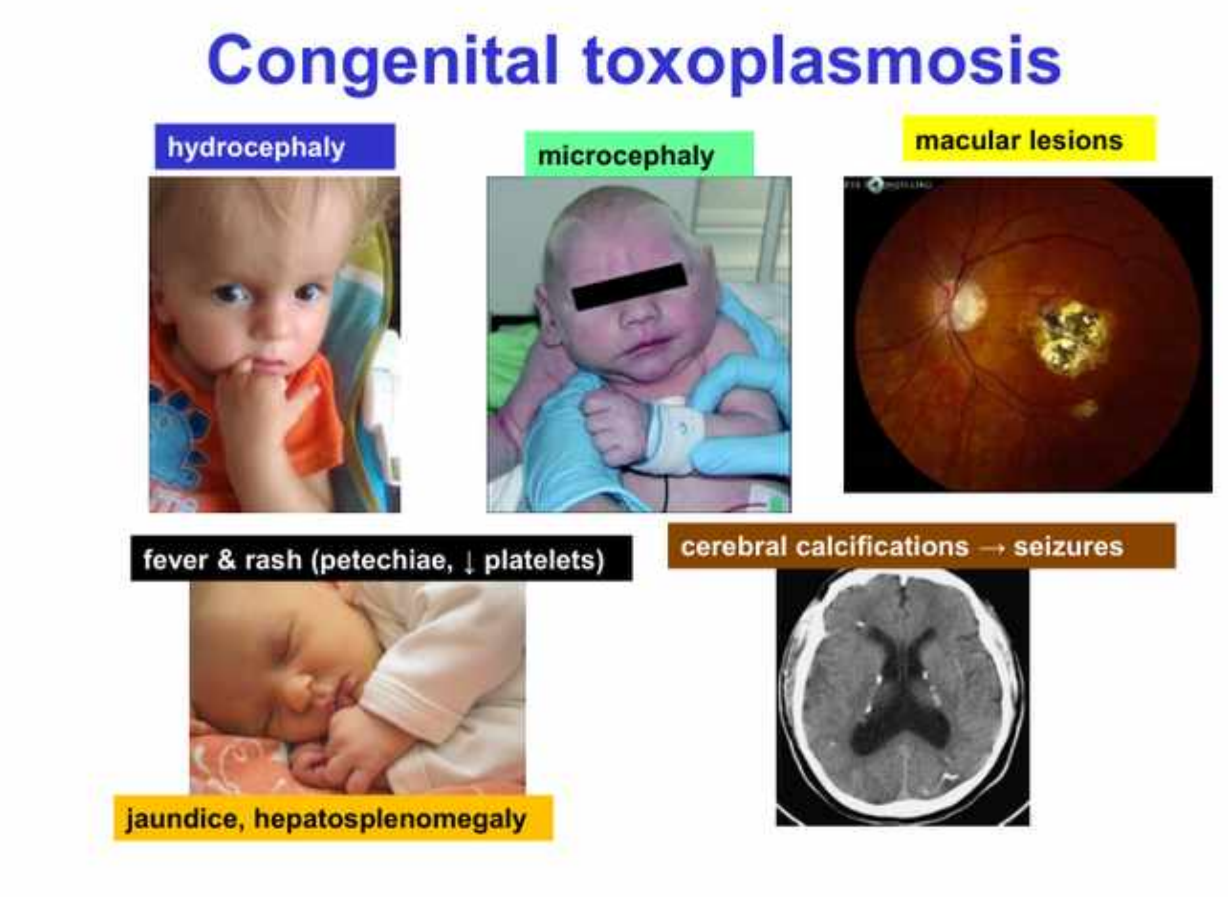
- Most infection is asymptomatic; some patients develop adenopathy/mononucleosis-like illness.
- Ocular toxoplasmosis: **chorioretinitis**, classically described as “headlights in a fog.”
- Immunocompromised hosts can develop **brain abscesses**.



Lecture slide 63 - full slide retained. The slide's classic phrase is 'headlights in a fog' for active chorioretinitis.

Slides 65-66: congenital toxoplasmosis

Fisher highlights hydrocephaly/microcephaly, macular lesions/chorioretinitis, cerebral calcifications with seizures, hepatosplenomegaly/jaundice, and thrombocytopenic petechiae.



Lecture slide 66 - full slide retained. The congenital slide is a visual high-yield set: hydrocephalus, cerebral calcifications, ocular lesions, and systemic findings.

CLASSIC CONGENITAL BOARD TRIAD

Chorioretinitis + hydrocephalus + intracranial calcifications.

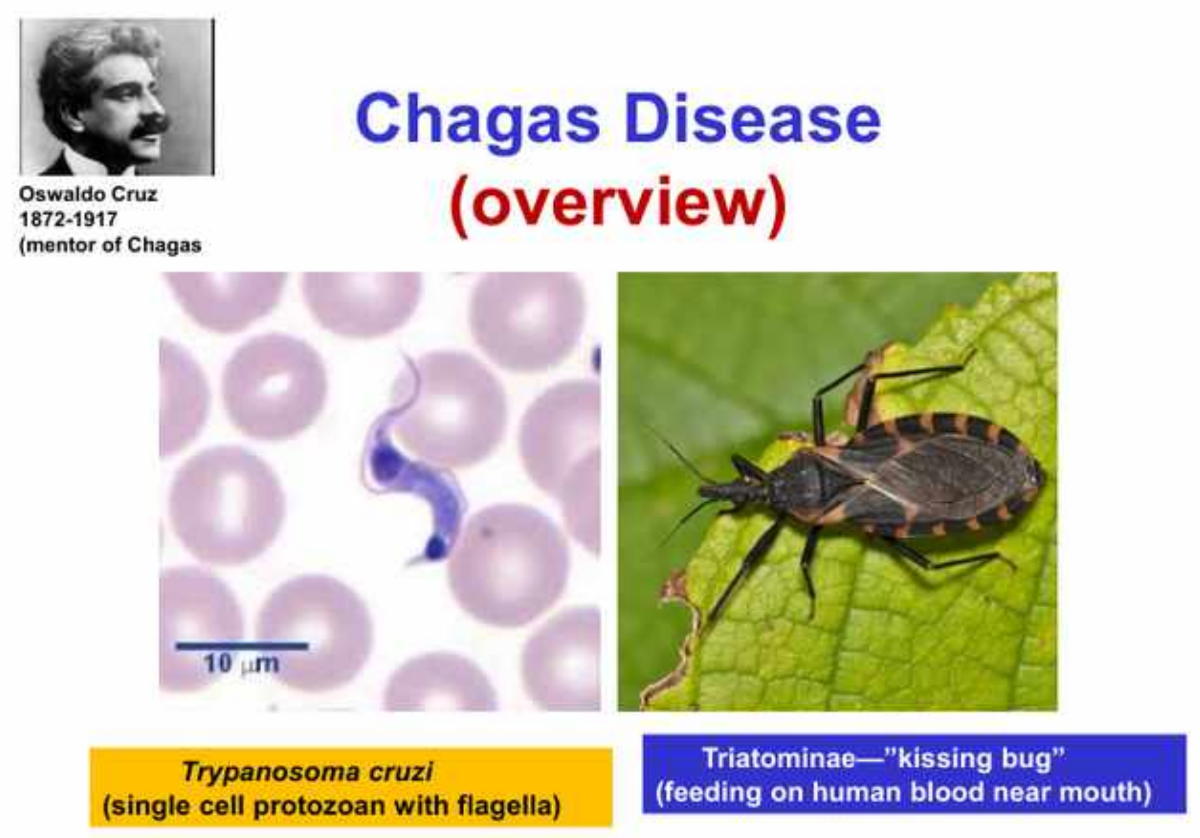
Slide 67 treatment in Fisher deck

- Self-limited uncomplicated infection: often no treatment.
- Severe/persistent disease: pyrimethamine + sulfadiazine.
- Congenital disease: prolonged pyrimethamine + sulfadiazine regimen in the slide.

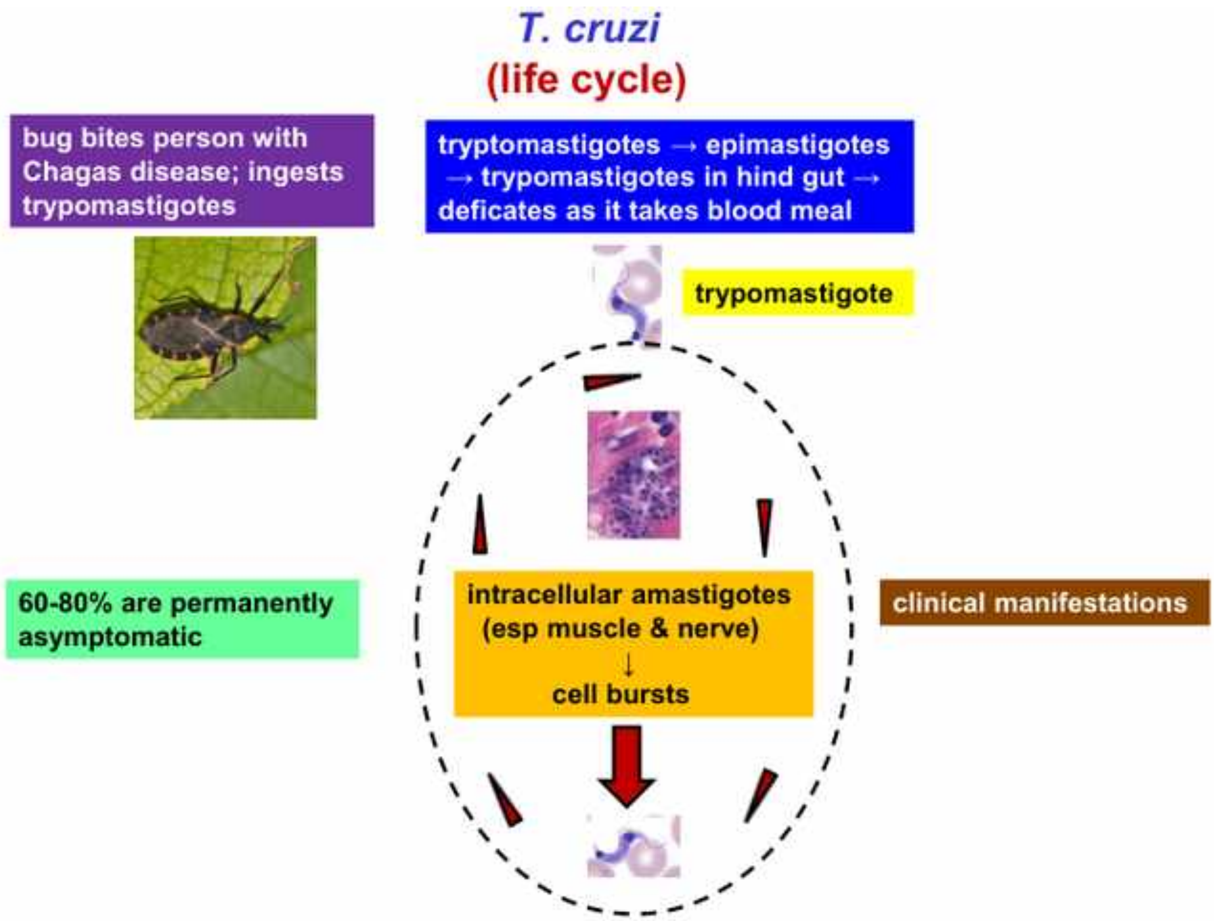
Slides 68-75

Chagas disease: T. cruzi from acute parasitemia to chronic heart/GI disease

Chagas LO



Lecture slide 69 - full slide retained. The vector is the triatomine 'kissing bug.' The parasite is Trypanosoma cruzi, a flagellated protozoan.



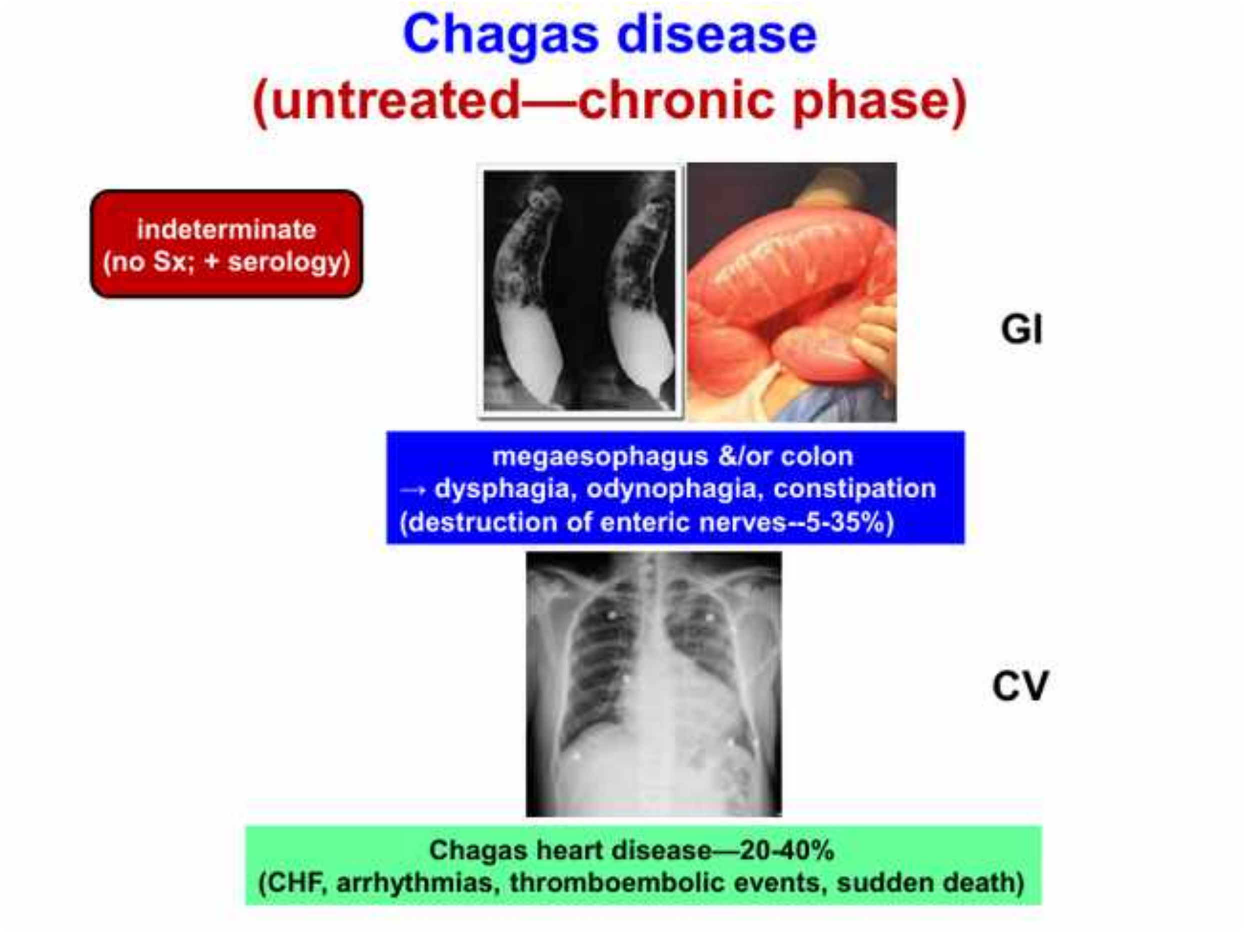
Lecture slide 70 - full slide retained. The bug becomes infected by ingesting trypomastigotes, then later defecates infectious forms during another blood meal. In humans, intracellular amastigotes preferentially involve muscle and nerve tissues; infected cells eventually rupture.

TRANSMISSION TRAP

The bite itself is not the key inoculum. The triatomine bug **defecates near the bite**; contaminated feces enter broken skin or mucosa.

Slide 71: acute phase

- Trypomastigotes circulate in blood.
- Local inoculation lesion = **chagoma**.
- Periorbital inoculation/edema = **Romaña sign**.
- Possible acute myocarditis, meningoencephalitis, hepatosplenomegaly, atypical lymphocytosis.



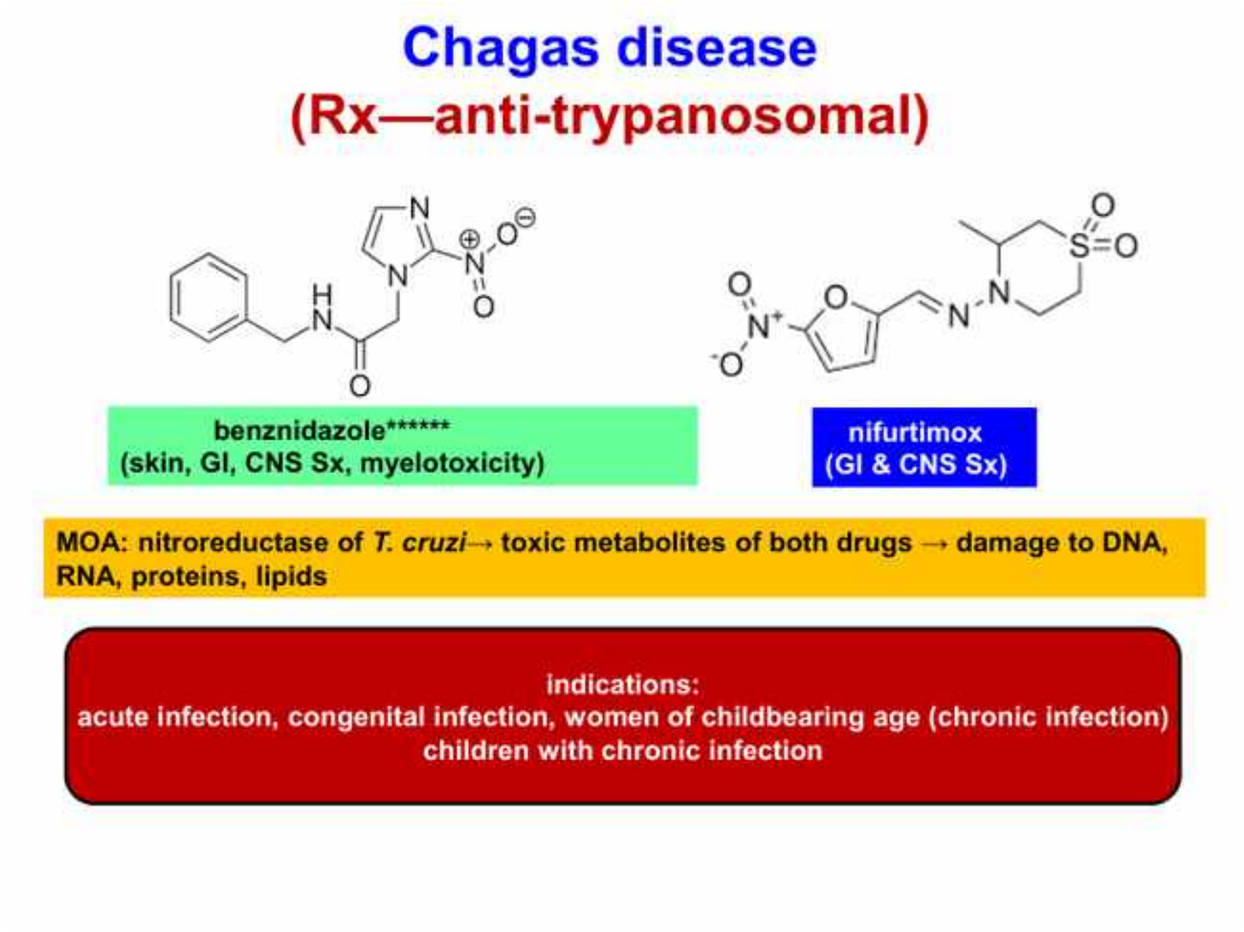
Lecture slide 72 - full slide retained. Untreated chronic disease may stay indeterminate or evolve into cardiomyopathy/arrhythmias/CHF/thromboembolism and GI denervation causing megaesophagus or megacolon.

CHRONIC CHAGAS COLD SET

Dilated/arrhythmogenic cardiomyopathy + megaesophagus/megacolon. Mechanistic theme: chronic tissue injury and destruction of muscle/enteric neural structures.

Slide 73: diagnosis

- **Acute:** microscopy or PCR.
- **Chronic:** serology (lecture shows T. cruzi antibody ELISA).



Lecture slide 74 - full slide retained. Fisher lists benznidazole and nifurtimox and highlights their toxic-metabolite mechanism through parasite nitroreductase. Treatment of established chronic cardiac/GI sequelae follows standard organ-specific care.

Phase	Diagnostic anchor	Treatment principle
Acute	Parasitemia -> microscopy/PCR	Antitrypanosomal therapy
Chronic indeterminate	Serology positive, no symptoms	Antitrypanosomal indications depend on age/context; Fisher highlights children and women of childbearing age among groups treated
Chronic organ disease	Cardiac/GI evaluation	Manage cardiomyopathy/arrhythmia and achalasia/ megacolon in addition to parasite-directed decisions

Slides 76-85 + synthesis **One-table parasite fingerprint map** Review

Disease	Transmission	Key tissue/ cell	Best clue	Treatment anchor in Fisher deck
Falciparum malaria	Anopheles mosquito	Liver -> any RBC	Crescent gametocyte, sequestration, severe disease	ACT if uncomplicated; IV artesunate if severe
Vivax/ovale malaria	Anopheles mosquito	Liver hypnozoite + reticulocyte	48-h fever; Schüffner-type changes; relapse	Blood-stage therapy + primaquine/ tafenoquine radical cure after G6PD assessment
P. malariae	Anopheles mosquito	Senescent RBC	72-h fever + band forms	Chloroquine-sensitive treatment in lecture framework
P. knowlesi	Mosquito; zoonotic primate reservoir	RBC	Daily cycle; malariae-like smear; may be severe	Treat as clinically appropriate malaria based on severity
Babesiosis	Ixodes tick / transfusion	RBC only	**Maltese cross**	Atovaquone + azithro; severe clinda + quinine
Toxoplasmosis	Cat feces / undercooked meat / congenital	Intracellular tissue infection	Chorioretinitis, brain abscess, congenital calcifications	Pyrimethamine + sulfadiazine when treatment indicated
Chagas	Triatomine fecal contamination	Muscle + nerves	Romaña/chagoma -> cardiomyopathy + megaesophagus/ megacolon	Benznidazole or nifurtimox

FISHER'S END-OF-LECTURE QUESTION ANSWERS

Falciparum: crescent gametocyte; schizonts rare peripherally; most common in sub-Saharan Africa.
Malariae: senescent RBC + q72 h fever.
Sexual malaria cycle: mosquito intestinal tract.
Babesia: Maltese cross.
Chagas: Brazilian patient with achalasia is the classic vignette.

Official-Learning-Objective Synthesis

Official objective	What you should be able to do without notes
Malaria life cycle	Trace sporozoite -> liver schizont -> merozoite -> RBC trophozoite/ schizont -> gametocyte -> mosquito sexual cycle.
Malaria pathogenesis	Connect RBC rupture to fever/hemolysis and falciparum cytoadherence/ sequestration to cerebral and multiorgan severe malaria.
Malaria diagnosis/ species	Recognize falciparum crescent gametocytes, malariae bands, vivax/ovale relapsing pattern, knowlesi daily cycle, and thick vs thin smear logic.
Malaria treatment/ prevention	Know ACT/uncomplicated falciparum, IV artesunate/severe disease, and vivax/ovale radical cure with G6PD caution; understand major drug mechanisms.
Babesiosis	Recognize Ixodes/transfusion + hemolysis + Maltese cross; distinguish from malaria; know atovaquone/azithro vs severe clinda/quinine.
Chagas	Trace fecal contamination -> trypomastigote/amastigote cycle; distinguish acute Romaña/chagoma from chronic cardiac/GI disease; know acute PCR/microscopy, chronic serology, benznidazole/nifurtimox.

LAST 60 SECONDS

Falciparum: any RBC, crescent gametocyte, sequestration, severe -> IV artesunate if severe.

Vivax/ovale: retics + hypnozoites -> relapse -> primaquine/tafenoquine after G6PD check.

Malariae: old RBC + band form + 72 h.

Knowlesi: daily + malariae-like + potentially severe.

Babesia: Ixodes + RBC only + Maltese cross.

Chagas: kissing-bug feces + intracellular amastigotes -> cardiomyopathy + megacolon/megaesophagus.

Source hierarchy: Fisher's 85-slide Protozoan Parasites deck is the primary one-to-one source. The uploaded “Lecture 10: Malaria” Word document supplements malaria only. Official Week 7 objectives prioritize malaria, babesiosis, and Chagas; toxoplasmosis is retained as lecture-deck content but flagged lower priority.

Ozzy checkpoint - click him after you can separate malaria from babesiosis in one sentence.